Module - 2

Continuous Probability Distributions

Continuous Random Variable (CRV)

A random variable X is said to continuous random variable if it takes all possible values in an interval.

Probability Density Function (Pdf)

A function $f(x)$ of the continuous random variable X is said to be a pdf if it satisfies the following conditions:

(i) $f(x) \geq 0$ for all $x \in R$

(ii) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

iii) $P[a < X < b] = \int_a^b f(x)\,dx$

Cumulative Distribution Function

If X is a continuous random variable with pdf $f(x)$, then its cumulative distribution function denoted by $F(x)$ is given by,

$$F(x) = P[X \leq x] = \int_{-\infty}^{x} f(x)\,dx.$$

Properties

* $P[a \leq x \leq b] = F(b) - F(a)$
* $F(-\infty) = 0$, $F(\infty) = 1$
* $F(b) \geq F(a)$ $\forall\, b \geq a$
* $f(x) = \frac{d}{dx} F(x)$ $\boxed{\text{OR}}$ $F(x) = \int_{-\infty}^{x} f(x)\,dx$

Q. A continuous random variable has the pdf $f(x) = \begin{cases} k e^{-2x}, & x > 0 \\ 0, & x \le 0 \end{cases}$

Then find (i) value of K (ii) $P[0 < X < 3]$ (iii) $P[X > 0.5]$ (iv) $F(x)$

Sol. (i) To find k

Since $f(x)$ is a pdf, $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$-\infty \quad 0 \quad \infty$

$e^{-\infty} = 0$

$e^{0} = 1$

$$\Rightarrow \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{\infty} f(x)\,dx = 1$$

$$\Rightarrow \int_{-\infty}^{0} 0 + \int_{0}^{\infty} k e^{-2x} = 1 \Rightarrow k\left[\frac{e^{-2x}}{-2}\right]_{0}^{\infty} = 1$$

$$\Rightarrow \frac{k}{-2}\left[e^{-\infty} - e^{0}\right] = 1$$

$$\Rightarrow \frac{k}{-2}(0-1) = 1 \quad \Rightarrow \underline{\underline{k = 2}}$$

(ii) To find $P[0 < X < 3]$

$$P[0<X<3] = \int_{0}^{3} f(x)\,dx = \int_{0}^{3} k e^{-2x}\,dx = k\left(\frac{e^{-2x}}{-2}\right)_{0}^{3} = \frac{k}{-2}\left[e^{-6} - e^{-0}\right]$$

$$= \frac{k}{-2}\left[e^{-6} - 1\right] = \frac{2}{-2}\left[e^{-6} - 1\right] = \underline{\underline{1 - e^{-6}}}$$

(iii) To find $P[X > 0.5]$

$$P[X > 0.5] = \int_{0.5}^{\infty} f(x)\,dx = \int_{0.5}^{\infty} k e^{-2x}\,dx = k\left(\frac{e^{-2x}}{-2}\right)_{0.5}^{\infty}$$

$$= \frac{k}{-2}\left[e^{-\infty} - e^{-2\times 0.5}\right] = \frac{k}{-2}\left[0 - e^{-1}\right] = \frac{2}{-2}\left(-e^{-1}\right) = \underline{\underline{e^{-1}}}$$

(iv) To find $F(x)$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{0}^{x} f(x)\,dx = \int_{0}^{x} k e^{-2x}\,dx = \left(k\frac{e^{-2x}}{-2}\right)_{0}^{x}$$

$$= \frac{k}{-2}\left[e^{-2x} - e^{-0}\right] = \frac{k}{-2}\left[e^{-2x} - 1\right] = \frac{2}{-2}\left[e^{-2x} - 1\right] = \underline{\underline{1 - e^{-2x}}}$$

Thus $F(x) = \begin{cases} 1 - e^{-2x}, & x > 0 \\ 0, & x \le 0 \end{cases}$

Q. Find c for which the following is a valid pdf of a continuous random variable: $f(x) = cxe^{-x}, \; 0 < x < \infty$

Sol: $\int_{-\infty}^{\infty} f(x)\,dx = 1 \Rightarrow \int_{-\infty}^{\infty} cxe^{-x}\,dx = 1$

$u = x \quad v = e^{-x}$

$u' = 1 \quad v_1 = \frac{e^{-x}}{-1}$

$u'' = 0$

$v_2 = \frac{e^{-x}}{(-1)^2}$

$\int uv = uv_1 - u'v_2 + \cdots$

$e^{-\infty} = 0$

$$\Rightarrow \int_0^{\infty} cxe^{-x}\,dx = 1$$

$$\Rightarrow c\left[x \cdot \frac{e^{-x}}{-1} - 1 \cdot \frac{e^{-x}}{1}\right]_0^{\infty} = 1$$

$$\Rightarrow c\left[(0-0) - (0 - e^{-0})\right] = 1$$

$$\Rightarrow \underline{\underline{c = 1}}$$

Q. Find the value of b so that the following function is a valid pdf. $f(x) = \begin{cases} 2x, & 0 \le x \le b \\ 0, & \text{otherwise} \end{cases}$. Also find $P[X \geq 0.5]$

Sol: $\int_{-\infty}^{\infty} f(x)\,dx = 1 \Rightarrow \int_0^b 2x\,dx = 1 \Rightarrow 2\left(\frac{x^2}{2}\right)_0^b = 1$

$$\Rightarrow \frac{2}{2}\left[b^2 - 0\right] = 1$$

$$\Rightarrow b^2 = 1$$

$$\Rightarrow b = \pm 1$$

$\left\{ \begin{array}{l} 0 \le x \le b \\ \Rightarrow b \geq 0 \end{array} \right.$

Here $\underline{\underline{b = +1}}$ since $0 \le x \le b$

<u>To find $P[X \geq 0.5]$</u>

$$P[X \geq 0.5] = \int_{0.5}^{\infty} f(x)\,dx = \int_{0.5}^{b} 2x\,dx = \int_{0.5}^{1} 2x\,dx = \left(2\frac{x^2}{2}\right)_{0.5}^{1}$$

$$= 1^2 - (0.5)^2 = 1 - 0.25 = \underline{\underline{0.75}}$$

[Q] A continuous random variable X has pdf $f(x) = \frac{k}{1+x^2}$, $-\infty < x < \infty$

(i) Find k (ii) Find $F(x)$ (iii) Find $P[X \geq 0]$

Sol. (i) To find k

Since $f(x)$ is a pdf, we have $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\Rightarrow \int_{-\infty}^{\infty} \frac{k}{1+x^2}\,dx = 1 \Rightarrow k\left[\tan^{-1}x\right]_{-\infty}^{\infty} = 1$

$\Rightarrow k\left[\tan^{-1}\infty - \tan^{-1}(-\infty)\right] = 1$

$\left\{\begin{array}{l}\tan\frac{\pi}{2} = \infty \\ \frac{\pi}{2} = \tan^{-1}(\infty)\end{array}\right.$

$\tan(-\theta) = -\tan\theta$

$\Rightarrow k\left[\frac{\pi}{2} - -\frac{\pi}{2}\right] = 1$

$\Rightarrow k \cdot 2\frac{\pi}{2} = 1 \Rightarrow \underline{\underline{k = 1/\pi}}$

(ii) To find $F(x)$

We know, $F(x) = P[X \leq x] = \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{x} \frac{k}{1+x^2}\,dx$

$= k\left[\tan^{-1}x\right]_{-\infty}^{x} = k\left[\tan^{-1}x - \tan^{-1}-\infty\right]$

$= k\left[\tan^{-1}x - -\frac{\pi}{2}\right] = \underline{\underline{\frac{1}{\pi}\left[\tan^{-1}x + \frac{\pi}{2}\right]}}$ $\because k = \frac{1}{\pi}$

(iii) To find $P[X \geq 0]$

$P[X \geq 0] = \int_{0}^{\infty} f(x)\,dx = \int_{0}^{\infty} \frac{k}{1+x^2}\,dx = k\left(\tan^{-1}x\right)_{0}^{\infty}$

$= k\left[\tan^{-1}\infty - \tan^{-1}0\right] = k\left[\frac{\pi}{2} - 0\right] = k\frac{\pi}{2} = \frac{1}{\pi}\cdot\frac{\pi}{2} = \underline{\underline{\frac{1}{2}}}$

Q. Diameter of an electric cable, say X, is assumed to be a continuous random variable with the function

$$f(x) = \begin{cases} 6x(1-x), & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) Check whether $f(x)$ is a pdf

(ii) Determine the value of b such that $P[X<b] = P[X>b]$

Sol: (i) To check whether $f(x)$ is a pdf

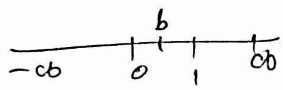
* $f(x) = 6x(1-x)$ is ≥ 0 for all $0 \le x \le 1$

* $\int_{-\infty}^{\infty} f(x)\,dx = \int_0^1 6x(1-x)\,dx = \int_0^1 (6x - 6x^2)\,dx$

$$= \left(6\frac{x^2}{2} - 6\frac{x^3}{3}\right)_0^1 = \frac{6}{2} - \frac{6}{3} = \underline{\underline{1}}$$

Since $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$ given $f(x)$ is a pdf

(ii) To find b



Given: $P[X<b] = P[X>b]$

$\Rightarrow \int_{-\infty}^{b} f(x)\,dx = \int_b^{\infty} f(x)\,dx$; Here $f(x) = \begin{cases} 6x(1-x), & [0,1] \\ 0, & \text{other} \end{cases}$

$\int_0^b 6x(1-x)\,dx = \int_b^1 6x(1-x)\,dx = \frac{1}{2}$ $\quad \because \int_0^1 f(x)\,dx = 1$

$\int_0^b (6x - 6x^2)\,dx = \frac{1}{2} \Rightarrow 6\left(\frac{x^2}{2} - \frac{x^3}{3}\right)_0^b = \frac{1}{2} \Rightarrow \frac{b^2}{2} - \frac{b^3}{3} = \frac{1}{12}$

$\therefore \frac{3b^2 - 2b^3}{6} = \frac{1}{12} \Rightarrow 3b^2 - 2b^3 = \frac{6}{12} \Rightarrow 3b^2 - 2b^3 = \frac{1}{2}$

$\Rightarrow 6b^2 - 4b^3 - 1 = 0 \Rightarrow b = \frac{1}{2}, \frac{1+\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}$

$\therefore \underline{\underline{b = \frac{1}{2}}}$ (other roots are not in $[0,1]$)

Q. If the pdf of a random variable is given by

$f(x) = \begin{cases} Kx^2, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$ then find (i) value of K

(ii) $P\left[\frac{1}{4} < X < \frac{1}{2}\right]$

(iii) $P\left[X > \frac{2}{3}\right]$

Sol: (i) To find K

Since f(x) is a pdf, we've $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\Rightarrow \int_0^1 f(x)\,dx = 1 \Rightarrow \int_0^1 kx^2\,dx = 1 \Rightarrow k\left(\frac{x^3}{3}\right)_0^1 = 1$

$\Rightarrow k\left(\frac{1}{3}\right) = 1 \Rightarrow \underline{k = 3}$

(ii) To find $P\left[\frac{1}{4} < X < \frac{1}{2}\right]$

$P\left[\frac{1}{4} < X < \frac{1}{2}\right] = \int_{1/4}^{1/2} f(x)\,dx = \int_{1/4}^{1/2} kx^2\,dx = 3\int_{1/4}^{1/2} x^2\,dx$

$= 3\left[\frac{x^3}{3}\right]_{1/4}^{1/2} = \left(\frac{1}{2}\right)^3 - \left(\frac{1}{4}\right)^3 = \frac{1}{8} - \frac{1}{64} = \underline{\frac{7}{64}}$

(iii) To find $P[X > 2/3]$

$P\left[X > \frac{2}{3}\right] = \int_{2/3}^{\infty} f(x)\,dx = \int_{2/3}^{1} kx^2\,dx = \int_{2/3}^{1} 3x^2\,dx$

$= 3\left(\frac{x^3}{3}\right)_{2/3}^{1} = 1^3 - \left(\frac{2}{3}\right)^3 = 1 - \frac{8}{27} = \underline{\frac{19}{27}}$

Q] If a continuous random variable X has the pdf
$f(x) = \begin{cases} K(1-x^2), & 0<x<1 \\ 0, & \text{elsewhere} \end{cases}$ then find (i) value of K
(ii) Distribution function
(iii) Using distribution function find $P[0.4 < X < 0.6]$

Sol:

(i) To find K

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \Rightarrow \int_0^1 K(1-x^2)\,dx = 1 \Rightarrow K\left[x - \frac{x^3}{3}\right]_0^1 = 1$$

$$\Rightarrow K\left[1-\frac{1}{3}\right] = 1 \Rightarrow K\left(\frac{2}{3}\right) = 1 \Rightarrow K = 3/2$$

(ii) To find distribution function (ie $F(x)$)

$$F(x) = \int_{-\infty}^{x} f(x)\,dx \quad \text{by definition}$$

$$= \int_0^x K(1-x^2)\,dx = \frac{3}{2}\left[x - \frac{x^3}{3}\right]_0^x = \frac{3}{2}\left[x - \frac{x^3}{3}\right]$$

(iii) To find $P[0.4 < X < 0.6]$ using $F(x)$

$$P[0.4 < X < 0.6] = F(0.6) - F(0.4)$$

$$= \frac{3}{2}\left[0.6 - \frac{0.6^3}{3}\right] - \frac{3}{2}\left[0.4 - \frac{0.4^3}{3}\right]$$

$$= 0.224$$

Q. Find the pdf of a random variable X if

$$F(x) = \begin{cases} 1 - \frac{1}{x^2}, & x > 1 \\ 0, & x \le 1 \end{cases}$$

. Also find $P[X<3]$

Sol: * $f(x) = \frac{d}{dx} F(x) = \frac{d}{dx}\left(1 - \frac{1}{x^2}\right) = \begin{cases} \frac{2}{x^3}, & x > 1 \\ 0, & x \le 1 \end{cases}$

* $P[X<3] = \int_{-\infty}^{3} f(x)\, dx$

$$= \int_{1}^{3} f(x)\, dx = \int_{1}^{3} \frac{2}{x^3} dx = 2\int_{1}^{3} x^{-3}\, dx = 2\left(\frac{x^{-3+1}}{-3+1}\right)_1^3$$

$$= 2\left(\frac{x^{-2}}{-2}\right)_1^3 = -\left(3^{-2} - 1^{-2}\right) = -\left(\frac{1}{9} - 1\right) = \underline{\underline{\frac{8}{9}}}$$

Q. If X is a continuous random variable with pdf

$$f(x) = \begin{cases} x/8, & 0 < x < 2 \\ 1/4, & 2 < x < 4 \\ 1/8(6-x), & 4 < x < 6 \\ 0, & \text{otherwise} \end{cases}$$

(i) Find the distribution function $F(x)$

(ii) verify $F'(x) = f(x)$

Sol: (i) Distribution function is, $F(x) = P[X \le x] = \int_{-\infty}^{x} f(x)\, dx$.

<u>when $0<x<2$</u>

$$F(x) = \int_{-\infty}^{x} f(x)\, dx = \int_{0}^{x} \frac{x}{8} dx = \frac{1}{8}\left(\frac{x^2}{2}\right)_0^x = \underline{\underline{\frac{x^2}{16}}}$$

<u>when $2<x<4$</u>

$$F(x) = \int_{-\infty}^{x} f(x)\, dx = \int_{0}^{2} f(x)\, dx + \int_{2}^{x} f(x)\, dx = \int_{0}^{2} \frac{x}{8} dx + \int_{2}^{x} \frac{1}{4} dx$$

Q. If X is a continuous random variable with pdf

$$f(x) = \begin{cases} x/8 & , \ 0 < x < 2 \\ 1/4 & , \ 2 < x < 4 \\ 1/8(6-x) & , \ 4 < x < 6 \\ 0 & , \ \text{otherwise} \end{cases}$$

(i) Find the distribution function $F(x)$

(ii) Verify $F'(x) = f(x)$

Sol: (i)

Distribution function is, $F(x) = P[X \le x] = \int_{-\infty}^{x} f(x)\,dx$.

When $0 < x < 2$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{0}^{x} \frac{x}{8}\,dx = \frac{1}{8}\left(\frac{x^2}{2}\right)_0^x = \underline{\underline{\frac{x^2}{16}}}$$

When $2 < x < 4$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{0}^{2} f(x)\,dx + \int_{2}^{x} f(x)\,dx = \int_{0}^{2} \frac{x}{8}\,dx + \int_{2}^{x} \frac{1}{4}\,dx$$

$$F(x) = \frac{1}{8}\left(\frac{x^2}{2}\right)_0^2 + \frac{1}{4}(x)_2^x$$

$$= \frac{1}{8}\left(\frac{4}{2} - 0\right) + \frac{1}{4}(x-2) = \frac{1}{4} + \frac{1}{4}x - \frac{2}{4} = \frac{x}{4} - \frac{1}{4} = \underline{\underline{\frac{x-1}{4}}}$$

<u>When $4 < x < 6$</u>

$$F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_0^2 f(x)\,dx + \int_2^4 f(x)\,dx + \int_4^x f(x)\,dx$$

$$= \int_0^2 \frac{x}{8}\,dx + \int_2^4 \frac{1}{4}\,dx + \int_4^x \frac{1}{8}(6-x)\,dx$$

$$= \frac{1}{8}\left(\frac{x^2}{2}\right)_0^2 + \frac{1}{4}(x)_2^4 + \frac{1}{8}\left(6x - \frac{x^2}{2}\right)_4^x$$

$$= \frac{1}{8}\left(\frac{2^2}{2}\right) + \frac{1}{4}(4-2) + \frac{1}{8}\left[\left(6x - \frac{x^2}{2}\right) - \left(24 - \frac{16}{2}\right)\right]$$

$$= \frac{1}{4} + \frac{2}{4} + \frac{1}{8}\left[6x - \frac{x^2}{2}\right] - \frac{24}{8} + \frac{16}{8\times 2}$$

$$= \frac{3}{4} + \frac{1}{8}\left[6x - \frac{x^2}{2}\right] - 3 + 1$$

$$= -\frac{5}{4} + \frac{6x}{8} - \frac{x^2}{16} = \underline{\underline{\frac{-20 + 12x - x^2}{16}}}$$

Thus, distribution function $F(x)$ is,

$$F(x) = \begin{cases} 0 & , x < 0 \\ x^2/16 & , 0 < x < 2 \\ \dfrac{x-1}{4} & , 2 < x < 4 \\ \dfrac{-20+12x-x^2}{16} & , 4 < x < 6 \\ 1 & , x > 6 \end{cases}$$

(ii) To verify $F'(x) = f(x)$

$$F'(x) = \begin{cases} 0 & , x < 0 \\ \dfrac{x}{8} & , 0 < x < 2 \\ \dfrac{1}{4} & , 2 < x < 4 \\ \dfrac{6-2x}{8} & , 4 < x < 6 \\ 0 & , x > 6 \end{cases}$$

$$= f(x).$$

Mean and Variance

$$\text{Mean} = \mu = E[X] = \int_{-\infty}^{\infty} x\, f(x)\,dx$$

$$\text{Variance} = V(X) = \sigma^2 = E[X^2] - [E(X)]^2$$

[Q] Let X be a random variable with pdf

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Find (i) Mean (ii) variance (iii) standard deviation (iv) $E[4X+5]$

Sol: (i) To find Mean

$$\text{Mean} = E[X] = \int_{-\infty}^{\infty} x\, f(x)\,dx$$

$$= \int_{-1}^{2} x \cdot \frac{x^2}{3}\,dx = \int_{-1}^{2} \frac{x^3}{3}\,dx = \frac{1}{3}\left[\frac{x^4}{4}\right]_{-1}^{2} = \frac{1}{12}\left[2^4 - (-1)^4\right] = \underline{\underline{\frac{15}{12}}}$$

(ii) To find Variance

$$\text{Variance} = E[X^2] - [E(X)]^2 \quad \text{--- ①}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_{-1}^{2} x^2 \cdot \frac{x^2}{3}\,dx = \int_{-1}^{2} \frac{x^4}{3}\,dx$$

$$= \frac{1}{3}\left(\frac{x^5}{5}\right)_{-1}^{2} = \frac{1}{3}\left[\frac{2^5}{5} - \frac{(-1)^5}{5}\right] = \frac{1}{15}\left[32 - -1\right] = \underline{\underline{\frac{33}{15}}}$$

From (i) $E(X) = \frac{15}{12}$. Substituting $E(X^2) = \frac{33}{15}$, $E(X) = \frac{15}{12}$

in ①, $\text{Variance} = \frac{33}{15} - \left(\frac{15}{12}\right)^2 = \underline{0.6375}$

(iii) To find standard deviation

Standard deviation $= \sigma = \sqrt{V(X)} = \sqrt{0.6375} = 0.8$

(iv) To find $E[4X+5]$

$E[4X+5] = 4E(X)+5$ $\qquad E(ax+b) = aE(x)+b$

$= 4 \times \frac{15}{12} + 5 = \underline{\underline{10}}$

Q If $f(x) = \begin{cases} k e^{-x/\sigma}, & 0 \le x < \infty \\ 0, & \text{elsewhere} \end{cases}$ is the pdf of a random variable, then show that mean and S.D are both $= \sigma$

Sol: Mean $= E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x k e^{-x/\sigma} dx$

$= k \int_0^{\infty} x e^{-x/\sigma} dx = k \left[x \cdot \frac{e^{-x/\sigma}}{-1/\sigma} - 1 \cdot \sigma^2 e^{-x/\sigma} \right]_{x=0}^{\infty}$

$= k \left[(0-0) - (0 - \sigma^2) \right] = \underline{\underline{k\sigma^2}}$

$u = x \quad v = e^{-x/\sigma}$
$u' = 1 \quad v_1 = \frac{e^{-x/\sigma}}{-1/\sigma}$
$u'' = 0$
$v_2 = \frac{e^{-x/\sigma}}{\frac{-1}{\sigma} \cdot \frac{-1}{\sigma}} = \sigma^2 e^{-x/\sigma}$

$\int uv = uv_1 - u'v_2 + \cdots$

$e^{-\infty} = 0$

S.D $= \sqrt{\text{variance}} = \sqrt{V(x)} = \sqrt{E(x^2) - [E(x)]^2}$

$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \cdot k e^{-x/\sigma} dx$

$= k \left[x^2 \cdot \frac{e^{-x/\sigma}}{-1/\sigma} - 2x \cdot \sigma^2 e^{-x/\sigma} + 2 \cdot \sigma^3 e^{-x/\sigma} \right]_{x=0}^{\infty}$

$= k \left[(0 - 0 + 0) - (0 - 0 - 2\sigma^3) \right] = \underline{\underline{+2k\sigma^3}}$

$u = x^2$
$u' = 2x$
$u'' = 2$
$u''' = 0$
$v_3 = \sigma^2 \cdot \frac{e^{-x/\sigma}}{-1/\sigma} = -\sigma^3 e^{-x/\sigma}$

$S.D = \sqrt{V(x)} = \sqrt{+2k\sigma^3 - (k\sigma^2)^2}$ — (1)

To find k: $\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_0^{\infty} k e^{-x/\sigma} = 1 \Rightarrow k \left[\frac{e^{-x/\sigma}}{-1/\sigma} \right]_0^{\infty} = 1$

$\Rightarrow -k\sigma [0 - 1] = 1 \Rightarrow k\sigma = 1 \Rightarrow k = 1/\sigma$

$\therefore$ (1) $\Rightarrow S.D = \sqrt{2 \cdot \frac{1}{\sigma}\sigma^3 - \frac{1}{\sigma^2}\sigma^4} = \sqrt{2\sigma^2 - \sigma^2} = \sqrt{\sigma^2} = \underline{\underline{\sigma}}$, Mean $= k\sigma^2 = \underline{\underline{\sigma}}$

Tutorial

Q. A continuous random variable X has $f(x) = \begin{cases} \frac{1}{2}x^2 e^{x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$

(i) Check whether $f(x)$ forms the pdf

(ii) If yes, find the mean and variance.

Home work

Q. If X be continuous random variable with pdf

$$f(x) = \begin{cases} 3x^2, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

find (i) $E(X^n)$

(ii) Mean

(iii) Variance.

Q. Find the mean and variance of a random variable X with the following density function.

$$f(x) = \begin{cases} K(10-x)x^2, & 0 \le x \le 1 \\ 0, & \text{elsewhere.} \end{cases}$$

Uniform Distribution

A continuous random variable X is said to follow uniform distribution if its pdf is,

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

[Q] Derive Mean of uniform distribution.

Sol: Mean $= E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx$

$$= \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{1}{b-a}\left[\frac{b^2}{2} - \frac{a^2}{2}\right] = \frac{b^2-a^2}{(b-a)2}$$

$$= \frac{\cancel{(b-a)}(b+a)}{\cancel{(b-a)}2} = \frac{a+b}{2}$$

[Q] Derive variance of uniform distribution.

Sol: Variance $= V(X) = E(X^2) - [E(X)]^2$ — ①

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\, dx = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a}\left(\frac{x^3}{3}\right)_a^b$$

$$= \frac{1}{3(b-a)}\left[b^3 - a^3\right] = \frac{1}{3(b-a)} \cdot (b-a)(b^2+ab+a^2)$$

$$= \frac{b^2+ab+a^2}{3}$$

Also, $E(X) = \frac{a+b}{2}$ (we proved)

Substituting the values of $E(x)$ + $E(x^2)$ in ①, we get

$$V(x) = \frac{b^2+ab+a^2}{3} - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{b^2+ab+a^2}{3} - \frac{(a^2+2ab+b^2)}{4}$$

$$= \frac{4b^2+4ab+4a^2-3a^2-6ab-3b^2}{12}$$

$$= \frac{b^2-2ab+a^2}{12} = \frac{(b-a)^2}{12}$$

NOTE : 1

Uniform distribution for a continuous random variable is also called a rectangular distribution

NOTE : 2

Cumulative distribution function of X in $a<x<b$ is

$$F(x) = P[X \le x] = \int_{-\infty}^{x} f(x)\,dx = \int_{a}^{x} \frac{1}{b-a}\,dx$$

$$= \frac{1}{b-a}\left[x\right]_a^x = \frac{1}{b-a}[x-a] = \frac{x-a}{b-a}$$

[Q] If X is uniformly distributed with mean 2 and variance 3. Find $P[X<1]$.

Sol: $P[X<1] = \int_{-\infty}^{1} f(x)\,dx$ —— ①   ; $X \to$ Unif-dist.

$\therefore f(x) = \frac{1}{b-a}$

$b = ?$ $a = ?$

Given: * Mean $= 2 \Rightarrow \frac{a+b}{2} = 2$

$\Rightarrow a+b = 4$ —— ②

* Variance $= 3 \Rightarrow \frac{(b-a)^2}{12} = 3$

$\Rightarrow (b-a)^2 = 36 \Rightarrow b-a = \pm 6$ —— ③

From ② & ③

$$\begin{aligned} a+b &= 4 \\ b-a &= 6 \\ \hline 2b &= 10 \end{aligned}$$

$b = 5$

$\therefore a = 4-5 = -1$

$a < b$ (here) $\boxed{a=-1,\ b=5 \mid \text{possible}}$ ✓

and

$$\begin{aligned} a+b &= 4 \\ b-a &= -6 \\ \hline 2b &= -2 \end{aligned}$$

$b = -1$

$a = 4-b = 4+1 = 5$

$a \nless b$ here $\boxed{a=5,\ b=-1 \mid \text{not possible}}$

Thus, $a=-1$, $b=5$ $\therefore f(x) = \frac{1}{5-1} = \frac{1}{6}$, $-1 < x < 5$

$\therefore$ ① becomes,

$P[X<1] = \int_{-\infty}^{1} f(x)\,dx = \int_{-1}^{1} \frac{1}{6}\,dx = \frac{1}{6}(x)_{-1}^{1}$

$= \frac{1}{6}[1 - -1] = \frac{2}{6} = \frac{1}{3}$

[Q] If X is uniformly distributed in $(-3,3)$, the find

(i) $P[X<0]$ (ii) $P[|X-1| \geq 1/2]$

Sol: $X \to$ unif. dist. in $(-3,3)$

$\therefore f(x) = \frac{1}{b-a} = \frac{1}{3-^-3} = \frac{1}{6}$ ie $f(x) = \frac{1}{6}$, $-3<x<3$

(i) To find $P[X<0]$

$$P[X<0] = \int_{-\infty}^{0} f(x)\,dx = \int_{-3}^{0} f(x)\,dx = \int_{-3}^{0} \frac{1}{6}\,dx = \frac{1}{6}[x]_{-3}^{0} = \cancel{\frac{1}{6}[\bar{3}]} =$$

$$= \frac{1}{6}[0-^-3] = \frac{3}{6} = \frac{1}{2}$$

(ii) To find $P[|X-1| \geq 1/2]$

$$P[|X-1| \geq 1/2] = 1 - P[|X-1| < \tfrac{1}{2}]$$

$$= 1 - P[-\tfrac{1}{2} < X-1 < \tfrac{1}{2}]$$

$$= 1 - P[\tfrac{1}{2} < X < \tfrac{3}{2}]$$

$$= 1 - \left[\int_{1/2}^{3/2} f(x)\,dx\right]$$

$$= 1 - \left[\int_{1/2}^{3/2} \frac{1}{6}\,dx\right]$$

$$= 1 - \left\{\frac{1}{6}[x]_{1/2}^{3/2}\right\} = 1 - \frac{1}{6}\left[\frac{3}{2} - \frac{1}{2}\right]$$

$$= 1 - \frac{1}{6} = 5/6$$

(9) If a random variable X is uniformly distributed with mean 1 and variance 3/4, find the prob. that the random variable assumes only negative values.

Sol: X → uniform distribution

$\boxed{P[X \text{ is } -ve] = P[X<0] = ? = \int_{-\infty}^{0} f(x)\,dx = ?}$

$f(x) = ?$

$\Rightarrow f(x) = \frac{1}{b-a}, \quad a<x<b$ —— (A)

Mean $=1 \Rightarrow \frac{a+b}{2} = 1 \Rightarrow a+b = 2$ —— (1)

Variance $= \frac{3}{4} \Rightarrow \frac{(b-a)^2}{12} = \frac{3}{4} \Rightarrow (b-a)^2 = 9$

$\Rightarrow b-a = \pm 3$ —— (2)

From (1) + (2):

$a+b=2$
$b-a=3$
$2b=5 \Rightarrow b = \frac{5}{2}$ $\quad \therefore a = 2-b = 2-\frac{5}{2} = \frac{-1}{2}$

also: $a+b=2$
$b-a=-3$
$2b=-1 \Rightarrow b=\frac{-1}{2}$ $\quad \therefore a = 2-b = 2+\frac{1}{2} = \frac{5}{2}$

$\therefore a=\frac{5}{2}, b=\frac{-1}{2}$ $\quad a \nless b$

Thus $a = -\frac{1}{2}, \quad b = \frac{5}{2}$

Hence (1) becomes,

$f(x) = \frac{1}{\frac{5}{2} - -\frac{1}{2}}, \quad -\frac{1}{2} < x < \frac{5}{2}$

$\boxed{* \ \frac{5}{2}+\frac{1}{2} = \frac{6}{2} = 3}$

ie $f(x) = \frac{1}{3}, \quad -\frac{1}{2} < x < \frac{5}{2}$.

Now, Prob $[X \text{ is } -ve] = P[X<0] = \int_{-\infty}^{0} f(x)\,dx$

$= \int_{-1/2}^{0} \frac{1}{3}\,dx = \frac{1}{3}[x]_{-1/2}^{0}$

$= \frac{1}{3}\left[0 - -\frac{1}{2}\right] = \frac{1}{6}$

[Q] Find the distribution function of random variable having a uniform distribution on (0,1).

Sol: Let X be a continuous random variable which follows uniform distribution on (0,1). Hence its pdf is

$$f(x) = \begin{cases} \frac{1}{1-0}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$* \quad f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$

Distribution function of X is F(x); where

$$F(x) = P[X \le x] = \int_{-\infty}^{x} f(x)\,dx = \int_{0}^{x} 1\,dx = x$$

Hence distribution function can be expressed as,

$$F(x) = \begin{cases} 0, & x < 0 \\ x, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

[Q] Find $P[|X-2| < 2]$ if X follows unif. dist. in (-3,3)

Sol: X → unif. dist. $\Rightarrow f(x) = \frac{1}{b-a}, \ a < x < b$.

Here: $(a,b) = (-3,3)$ $\therefore f(x) = \frac{1}{3-^-3} = \frac{1}{6}, \ -3 < x < 3$

Now, $P[|X-2| < 2] = P[-2 < X-2 < ^+2]$

$= P[0 < X < 4]$

$= \int_0^3 f(x)\,dx = \int_0^3 \frac{1}{6}\,dx = \frac{1}{6}[x]_0^3$

$= \frac{1}{6}[3] = \frac{1}{2}$ //

[Q] In certain experiments, the error made in determining the solubility of a substance is a random variable having the uniform density with $\alpha = -0.025$ and $\beta = 0.025$. What are the probabilities that such an error will be (i) between 0.010 & 0.015
(ii) between -0.012 & 0.012

Sol: $X \rightarrow$ uniform dist. with $\alpha = -0.025$, $\beta = 0.025$.

$\therefore f(x) = \dfrac{1}{\beta - \alpha} = \dfrac{1}{0.025 - (-0.025)} = \dfrac{1}{0.05}$, $-0.025 < x < 0.025$

(i) b/n 0.010 & 0.015

$$P[0.010 < X < 0.015] = \int_{0.010}^{0.015} f(x)\,dx$$

$$= \int_{0.010}^{0.015} \frac{1}{0.05}\,dx = \frac{1}{0.05}\left[x\right]_{0.010}^{0.015} = \frac{1}{0.05}\left[0.015 - 0.010\right]$$

$$= 0.01$$

(ii) b/n -0.012 & 0.012

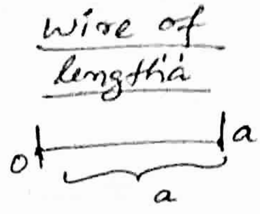
$$P[-0.012 < X < 0.012] = \int_{-0.012}^{0.012} f(x)\,dx$$

$$= \int_{-0.012}^{0.012} \frac{1}{0.05}\,dx = \frac{1}{0.05}\left(x\right)_{-0.012}^{0.012} = \frac{1}{0.05}\left[0.012 - (-0.012)\right]$$

$$= \frac{0.024}{0.05} = \frac{24}{50}$$

Q A wire of length 'a' unit is divided into two parts. If the part of random length X is uniformly distributed, then find (i) Mean
(ii) Variance
(iii) $E[X(a-X)]$

Sol: $X \to$ unif. distrib. $\Rightarrow f(x) = \dfrac{1}{b-a}$



$$\Rightarrow f(x) = \frac{1}{a-0} = \frac{1}{a}, \quad 0 < x < a$$

(i) <u>To find Mean</u>

$$\text{Mean} = E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$= \int_0^a x \cdot \frac{1}{a}\,dx = \frac{1}{a}\left(\frac{x^2}{2}\right)_0^a = \frac{1}{a}\left[\frac{a^2}{2} - 0\right] = \underline{\frac{a}{2}}$$

(ii) <u>To find variance</u>

$$\text{Var} X = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_0^a x^2 \cdot \frac{1}{a}\,dx = \frac{1}{a}\cdot\left(\frac{x^3}{3}\right)_0^a = \frac{1}{3a}(a^3 - 0)$$

$$= \underline{\underline{\frac{a^2}{3}}}$$

$$\therefore \text{Var} X = \frac{a^2}{3} - \left(\frac{a}{2}\right)^2 = \frac{a^2}{3} - \frac{a^2}{4} = \underline{\underline{\frac{a^2}{12}}}$$

(iii) <u>To find $E[X(a-X)]$</u>

$$E[X(a-X)] = \cancel{E[aX]} - = E[Xa - X^2]$$

$$= aE(X) - E(X^2)$$

$$= a\left(\frac{a}{2}\right) - \frac{a^2}{3} = \frac{a^2}{2} - \frac{a^2}{3} = \frac{3a^2 - 2a^2}{6} = \underline{\underline{\frac{a^2}{6}}}$$

[Q] A random variable X is uniformly distributed in $(-\alpha, \alpha)$, $\alpha > 0$. Find the value of α if $P[|X|<1] = P[|X|>1]$

Sol: $X \to$ unif. dist. [OR] we write $X \sim U(-\alpha, \alpha)$

$\therefore f(x) = \frac{1}{b-a}$ ie $f(x) = \frac{1}{\alpha - -\alpha} = \frac{1}{2\alpha}$, $-\alpha < x < \alpha$

Given: $P[|X|<1] = P[|X|>1]$

Since total probability $= 1$, we have, both $= 1/2$

Now, $P[|X|<1] = \frac{1}{2} \Rightarrow P[-1 < X < 1] = \frac{1}{2}$

$$\Rightarrow \int_{-1}^{1} f(x)\,dx = \frac{1}{2} \Rightarrow \int_{-1}^{1} \frac{1}{2\alpha}\,dx = \frac{1}{2}$$

$$\Rightarrow \frac{1}{2\alpha}(x)\Big|_{-1}^{1} = \frac{1}{2} \Rightarrow \frac{1}{2\alpha}[1 - -1] = \frac{1}{2}$$

$$\therefore \frac{2}{2\alpha} = \frac{1}{2} \Rightarrow \frac{1}{\alpha} = \frac{1}{2} \Rightarrow \underline{\underline{\alpha = 2}}$$

[Q] A random variable X has pdf, $f(x) = \begin{cases} \frac{1}{4}, & |x| < 2 \\ 0, & \text{otherwise} \end{cases}$

Find (i) $P[X<1]$ (ii) $P[|X|>1]$ (iii) $P[2X+3>5]$

Sol: Given, pdf is $f(x) = \frac{1}{4}$, $|x| < 2$

ie $f(x) = \frac{1}{4}$, $-2 < x < 2$

ie $f(x) = \frac{1}{2 - -2}$, $\underset{\downarrow\, a}{-2} < x < \underset{\downarrow\, b}{2}$

$\Downarrow \frac{1}{b-a}$

Thus, from the given pdf it follows that $X \sim U(-2,2)$

(i) To find $P[X<1]$

$$f(x) = \frac{1}{4}, \quad -2<x<2$$

$$P[X<1] = \int_{-\infty}^{1} f(x)\,dx = \int_{-2}^{1} \frac{1}{4}\,dx = \frac{1}{4}(x)_{-2}^{1} = \frac{1}{4}[1+2] = \frac{3}{4}$$

(ii) To find $P[|X|>1]$

$$P[|X|>1] = 1 - P[|X|<1]$$

$$= 1 - P[-1<X<1]$$

$$= 1 - \int_{-1}^{1} f(x)\,dx$$

$$= 1 - \int_{-1}^{1} \frac{1}{4}\,dx = 1 - \frac{1}{4}(x)_{-1}^{1} = 1 - \frac{1}{4}(1+1)$$

$$= 1 - \frac{1}{4} \times 2 = 1 - \frac{1}{2} = \frac{1}{2}$$

(iii) To find $P[2X+3>5]$

$$P[2X+3>5] = P[2X>2] = P[X>1]$$

$$= \int_{1}^{2} f(x)\,dx = \int_{1}^{2} \frac{1}{4}\,dx = \frac{1}{4}(x)_{1}^{2} = \frac{1}{4}$$

Home Work

19 A string of length 10m is divided into 2 parts. If the part of random length X is uniformly distributed, then find (i) $E(3X+2)$ (ii) $Var[2X+3]$ (iii) $E[X(10-X)]$

Q The number of cellphones sold daily at cell world is uniformly distributed with a minimum of 50 phones and a maximum of 100 phones. Find the probability that

(i) cell world will sell at least 75 phones per day.

(ii) the daily sales will fall between 70 and 80

Sol. X = no. of cell phones sold daily.

Given: X follows unif. dist. with min. value 50 and max. value 100.

ie $X \sim U(50, 100)$

$\therefore$ $f(x) = \frac{1}{b-a}$, $a < x < b$ becomes,

$$f(x) = \frac{1}{100-50} = \frac{1}{50}, \quad 50 < x < 100$$

(i) Prob[selling atleast 75 phones] $= P[X \geq 75]$

$$= \int_{75}^{100} f(x)\,dx$$

$$= \int_{75}^{100} \frac{1}{50}\,dx = \frac{1}{50}\left[x\right]_{75}^{100} = \frac{1}{50}\left[100-75\right] = \frac{25}{50} = \frac{1}{2}$$

(ii) Prob [sales fall b/n 70 & 80] $= P[70 < X < 80]$

$$= \int_{70}^{80} f(x)\,dx = \int_{70}^{80} \frac{1}{50}\,dx = \frac{1}{50}(x)_{70}^{80}$$

$$= \frac{1}{50}(80-70) = \frac{10}{50} = \frac{1}{5}$$

[Q] The cumilative distribution of a random variable X is given by,

$$F(x) = \begin{cases} 0 & \text{if } x < -\pi \\ \frac{x+\pi}{2\pi} & \text{if } -\pi < x < \pi \\ 1 & \text{if } x \geq \pi \end{cases}$$

(i) Find pdf, $f(x)$

(ii) Find $P\left[\frac{-\pi}{2} < X < \frac{\pi}{2}\right]$

(iii) show that X is uniformly distributed over $[-\pi, \pi]$

Sol: Given, $F(x)$.

(i) <u>To find pdf $f(x)$</u>

$$\text{pdf} - f(x) = \frac{d}{dx} F(x) = \frac{d}{dx}\left(\frac{x+\pi}{2\pi}\right) = \frac{1}{2\pi}.$$

Thus, $f(x) = \begin{cases} \frac{1}{2\pi}, & -\pi < x < \pi \\ 0, & \text{otherwise} \end{cases}$

(ii) <u>To find $P\left[\frac{-\pi}{2} < X < \frac{\pi}{2}\right]$</u>

$$P\left[\frac{-\pi}{2} < X < \frac{\pi}{2}\right] = \int_{-\pi/2}^{\pi/2} f(x)\, dx = \int_{-\pi/2}^{\pi/2} \frac{1}{2\pi} dx = \frac{1}{2\pi}\left[\frac{\pi}{2} - \frac{-\pi}{2}\right] = \frac{1}{2\pi} \times \pi = \frac{1}{2}$$

(iii) <u>To S.T X is uniformly distributed</u>

Here; $f(x) = \begin{cases} \frac{1}{2\pi}, & -\pi < x < \pi \\ 0, & \text{otherwise} \end{cases}$ which is the pdf of uniform distribution, since

pdf of unif. dist. in general is $f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$

Here $a = -\pi$, $b = \pi$ so that $\frac{1}{b-a} = \frac{1}{\pi - -\pi} = \frac{1}{2\pi}$

<u>Home Works</u>

[Q] If X is uniformly distributed in $(-\alpha, \alpha)$ with $\alpha > 0$ then

(i) find α such that $P[X < 1] = 1/3$

(ii) find α such that $P[X < \frac{1}{2}] = 0.7$

[Q] Calulate the mean and variance of for the rectangular distribution in $(0, 5)$

Exponential Distribution

A continuous ~~dist~~ random variable X is said to follow exponential distribution if its pdf is

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$\boxed{Q}$ Derive Mean and Variance of exponential distribution.

Sol: To find Mean:

$$\text{Mean} = E(X) = \int_{-\infty}^{\infty} x f(x)\,dx = \int_{0}^{\infty} x\, \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} x e^{-\lambda x} dx$$

$$= \lambda \left[x \cdot \frac{e^{-\lambda x}}{-\lambda} - 1 \cdot \frac{e^{-\lambda x}}{\lambda^2} \right]_{x=0}^{\infty}$$

$$= \lambda \left[(0-0) - \left(0 - \frac{1}{\lambda^2}\right) \right] = \lambda \left[\frac{1}{\lambda^2}\right]$$

$$= \underline{\underline{1/\lambda}}$$

$u = x, \quad v = e^{-\lambda x}$

$u' = 1, \quad v_1 = \frac{e^{-\lambda x}}{-\lambda}$

$u'' = 0, \quad v_2 = \frac{e^{-\lambda x}}{-\lambda \cdot \lambda}$

$\int uv = uv_1 - u'v_2 + \cdots$

To find variance

$$\text{Var} = V(X) = E(X^2) - [E(X)]^2$$

Now, $$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_0 x^2 . \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx$$

$$= \lambda \left[x^2 \cdot \frac{e^{-\lambda x}}{-\lambda} - 2x \cdot \frac{e^{-\lambda x}}{\lambda^2} + 2 \cdot \frac{e^{-\lambda x}}{-\lambda^3} \right]_{x=0}^{\infty}$$

$$= \lambda \left[(0 - 0 + 0) - \left(0 - 0 + 2 \cdot \frac{1}{-\lambda^3}\right) \right]$$

$$= \lambda \left(\frac{2}{\lambda^3}\right) = \frac{2}{\lambda^2}$$

$$\therefore V(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2} \;//$$

$u = x^2 \quad v = e^{-\lambda x}$

$u' = 2x \quad v_1 = \frac{e^{-\lambda x}}{-\lambda}$

$u'' = 2 \quad v_2 = \frac{e^{-\lambda x}}{-\lambda \cdot \lambda}$

$u''' = 0 \quad v_3 = \frac{e^{-\lambda x}}{(-\lambda)^3}$

Q A random variable follows exponential distribution with pdf $f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$. Find the probability that it will take on a value (i) between 1 & 3 (ii) greater than 0.5. Also find mean, variance & distribution function.

Sol: (i) $P[1<x<3] = \int_1^3 f(x)dx = \int_1^3 2e^{-2x}dx = 2\left(\frac{e^{-2x}}{-2}\right)_1^3$

$= \frac{2}{-2}\left(e^{-6} - e^{-2}\right) = \underline{e^{-2} - e^{-6}} \approx \underline{0.133}$

(ii) $P[X>0.5] = \int_{0.5}^{\infty} f(x)dx = \int_{0.5}^{\infty} 2e^{-2x}dx = 2\left[\frac{e^{-2x}}{-2}\right]_{0.5}^{\infty}$

$= \frac{2}{-2}\left[e^{-\infty} - e^{-2\times 0.5}\right] = -1\left[0 - e^{-1}\right] = e^{-1} = \underline{\underline{0.368}}$

Mean $= E(X) = \int_{-\infty}^{\infty} xf(x)dx = \int_0^{\infty} x \cdot 2e^{-2x}dx = 2\left[x \cdot \frac{e^{-2x}}{-2} - 1 \cdot \frac{e^{-2x}}{(-2)^2}\right]_0^{\infty}$

$= 2\left[(0-0) - \left(0 - \frac{1}{4}\right)\right] = 2\left(\frac{1}{4}\right) = \underline{\underline{\frac{1}{2}}}$

Variance $= E(x^2) - [E(x)]^2$

$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^{\infty} x^2 \cdot 2e^{-2x}dx = 2\int_0^{\infty} x^2 e^{-2x}dx$

$= 2\left[x^2 \cdot \frac{e^{-2x}}{-2} - 2x \cdot \frac{e^{-2x}}{(-2)^2} + 2 \cdot \frac{e^{-2x}}{(-2)^3}\right]_0^{\infty} = 2\left[(0-0+0) - \left(0 - 0 + 2 \cdot \frac{1}{-8}\right)\right]$

$= \frac{4}{8} = \underline{\underline{\frac{1}{2}}}$

$\therefore$ Variance $= \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{2} - \frac{1}{4} = \underline{\underline{\frac{1}{4}}}$

<u>Distribution function</u> $= F(x) = \int_{-\infty}^{x} f(x)dx = \int_0^{x} 2e^{-2x}dx$

$= 2\left(\frac{e^{-2x}}{-2}\right)_0^{x} = -1\left[e^{-2x} - 1\right]$

$= 1 - e^{-2x}$

[Q] If X follows an exponential distribution with $P[X \le 1] = P[X > 1]$, find the mean and variance of X

Sol: Given : X follows exp. dist.

$$\Rightarrow f(x) = \lambda e^{-\lambda x}, \quad x \geq 0 .$$

Also given that, $P[X \le 1] = P[X > 1]$.

Since total probability $= 1$, each of the above prob. $= \frac{1}{2}$

~~Let~~ Now we take $P[X \le 1] = \frac{1}{2}$

ie $$\int_{-\infty}^{1} f(x)\,dx = \frac{1}{2}$$

$$\Rightarrow \int_{0}^{1} \lambda e^{-\lambda x} = \frac{1}{2} \quad [\because x \geq 0 \text{ for exp. dist.}]$$

$$\Rightarrow \lambda \left[\frac{e^{-\lambda x}}{-\lambda}\right]_0^1 = \frac{1}{2}$$

$$\Rightarrow -1\left[e^{-\lambda \cdot 1} - e^{-\lambda \cdot 0}\right] = \frac{1}{2}$$

$$\Rightarrow -1\left[e^{-\lambda} - e^{0}\right] = \frac{1}{2}$$

$$\Rightarrow -1\left[e^{-\lambda} - 1\right] = \frac{1}{2}$$

$$\Rightarrow -e^{-\lambda} + 1 = \frac{1}{2}$$

$\Rightarrow \frac{1}{2} = e^{-\lambda}$ $\therefore$ Taking log on both sides,

$\log\left(\frac{1}{2}\right) = \log e^{-\lambda}$

$\Rightarrow \log 1 - \log 2 = -\lambda$

$\Rightarrow \quad 0 - \log 2 = -\lambda$

$\Rightarrow \quad \log 2 = \lambda$

Mean $= \frac{1}{\lambda} = \frac{1}{\log 2}$

Variance $= \frac{1}{\lambda^2} = \frac{1}{(\log 2)^2}$

Q The life time (in years) of an electronic component is an exponential random variable with mean 1 year. Find the lifetime L which a typical component is 60% certain to exceed.

Sol: Let X denote life time of an electronic component.

Given: X ~ exp. dist. Then $f(x) = \lambda e^{-\lambda x}, x \geq 0$.

Also, given that mean = 1 $\Rightarrow \frac{1}{\lambda} = 1 \Rightarrow \lambda = 1$

$\therefore f(x) = e^{-x}, x \geq 0$

Here, we need to find 'L' such that $P[X>L] = 60\%$

ie $P[X>L] = 0.60$

$\Rightarrow \int_L^{\infty} f(x)\,dx = 0.60 \Rightarrow \int_L^{\infty} e^{-x}\,dx = 0.60 \Rightarrow \left(\frac{e^{-x}}{-1}\right)_L^{\infty} = 0.60$

$\Rightarrow -\left[e^{-\infty} - e^{-L}\right] = 0.6 \Rightarrow -\left[0 - e^{-L}\right] = 0.6 \Rightarrow e^{-L} = 0.6$

$\Rightarrow \frac{1}{e^L} = 0.6 \Rightarrow \log 1 - \log e^L = \log 0.6 \Rightarrow -L = \log 0.6 \Rightarrow L = -\log_e(0.6)$

Q. The life time (in years) of an electronic component is an exponential random variable with mean 1 year. Find the lifetime L which a typical component is 60% certain to exceed.

Sol: Let X denote life time of an electronic component.

Given : X ~ exp. dist. Then $f(x) = \lambda e^{-\lambda x}, x \geq 0$.

Also, given that mean = 1 $\Rightarrow \frac{1}{\lambda} = 1 \Rightarrow \underline{\underline{\lambda = 1}}$

$\therefore f(x) = e^{-x}, x \geq 0$

Here, we need to find 'L' such that $P[X>L] = 60\%$

ie $P[X>L] = 0.60$

$\Rightarrow \int_L^\infty f(x)dx = 0.60 \Rightarrow \int_L^\infty e^{-x}dx = 0.60 \Rightarrow \left(\frac{e^{-x}}{-1}\right)_L^\infty = 0.60$

$\Rightarrow -[e^{-\infty} - e^{-L}] = 0.6 \Rightarrow -[0 - e^{-L}] = 0.6 \Rightarrow e^{-L} = 0.6$

$\Rightarrow \frac{1}{e^L} = 0.6 \Rightarrow \log 1 - \log e^L = \log 0.6 \Rightarrow -L = \log 0.6 \Rightarrow \underline{L = -\log_e(0.6)}$

Q] The time required to repair a machine is exponentially distributed with mean 2. What is the probability that a repair time exceeds 4 hours?

Sol: Let X denote repairing time.

Given: X → Exponential distribution.

∴ $f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$ —— ①

Given: Mean = 2; For an expon. dist. mean $= \frac{1}{\lambda}$

$\Rightarrow \frac{1}{\lambda} = 2 \Rightarrow \lambda = \frac{1}{2}$. ∴ ① becomes,

$f(x) = \frac{1}{2} e^{-\frac{1}{2}x}, \quad x \geq 0$

Now, prob. [repairing time exceeds 4 hours]

$= P[X > 4] = \int_4^\infty f(x)\,dx = \int_4^\infty \frac{1}{2} e^{-1/2 x}\,dx$

$= \frac{1}{2}\left[\frac{e^{-1/2 x}}{-1/2}\right]_4^\infty = -1\left[e^{-\infty} - e^{-4/2}\right]$

$= -1\left[0 - e^{-2}\right] = \underline{\underline{e^{-2}}}$

2] The length of time for one individual to be served at a cafteria is a r.v. having an exponential distribution with a mean of 3 minute. What is the prob. that he is served in less than 3 minutes.

1. X → expon. distr. $\Rightarrow f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$ —— ①

Given: mean = 3 $\Rightarrow \frac{1}{\lambda} = 3 \Rightarrow \lambda = \frac{1}{3}$ ∴ $f(x) = \frac{1}{3} e^{-1/3 x}, \quad x \geq 0$

Now; Prob [less than 3] $= P[X<3] = \int_{-\infty}^{3} f(x)\,dx = \int_0^3 \frac{1}{3} e^{-1/3 x}\,dx = \frac{1}{3}\left[\frac{e^{-1/3 x}}{-1/3}\right]_0^3$

$= -\left[e^{-1} - e^{0}\right] = \underline{1 - e^{-1}}$

Q. Let X denote the time between detections of a particle using a counter, and assume that X has an exponential distribution with mean 2 minutes per detection. Find the probability that we detect a particle within 30 seconds of starting the counter.

Sol. X → expon. distribution ∴ $f(x) = \lambda e^{-\lambda x}, x \geq 0$.

Given; mean = 2 minutes $\Rightarrow \frac{1}{\lambda} = 2 \Rightarrow \lambda = \frac{1}{2}$

∴ $f(x) = \frac{1}{2} e^{-\frac{1}{2}x}, \quad x \geq 0$.

Now,

$$P[\text{within 30 seconds}] = P[X \leq 30 \text{ seconds}]$$

$$= P[X \leq 0.5 \text{ minutes}]$$

$$= \int_{-\infty}^{0.5} f(x)\,dx = \int_{0}^{0.5} \frac{1}{2} e^{-\frac{1}{2}x}\,dx$$

$$= \frac{1}{2}\left[\frac{e^{-\frac{1}{2}x}}{-1/2}\right]_0^{0.5} = -1\left[e^{-\frac{0.5}{2}} - e^{0}\right] = \underline{\underline{1 - e^{-\frac{0.5}{2}}}}$$

Home Work

Q. If the distance that a car can run before its battery down is exponentially distributed with an average value of 5,000 km. If the owner desires to take a 2000 km trip, what is the probability that he will be able to complete his trip without having to replace the car battery.

Normal Distribution

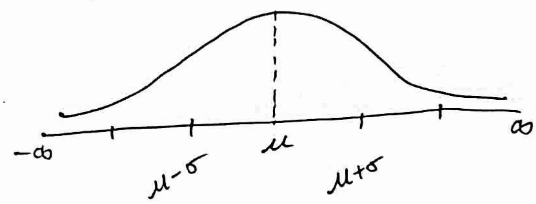
A continuous random variable X is said to follow normal distribution if its pdf is,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad , \quad -\infty < x < \infty$$

The normal distribution is also called Gaussian distribution

NOTE:

- The graph of normal distribution is shaped like the cross-section of a bell.
- The bell-shaped curve is symmetrical about mean μ.



Properties of Normal distribution:

* The normal curve is bell shaped and is symmetrical about the line $x = \mu$
* Mean, Median & Mode of normal distribution coincide
* Maximum probability occuring at the point $x = \mu$ is $\frac{1}{\sigma\sqrt{2\pi}}$
* The total area under the normal curve is 1.
  ∴ Area of the normal curve from $-\infty$ to μ is $\frac{1}{2}$ and from μ to ∞ is also $\frac{1}{2}$.
* Mean of normal distribution is : $E(X) = \mu$ and variance of " " is σ^2

Standard Normal distribution :

The normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$ is called standard normal distribution. Its pdf is,

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \quad , \quad -\infty < z < \infty$$

It is denoted by $Z \sim N(0,1)$

Cumulative distribution function of Standard Normal Variable :-

$$F(z) = \int_{-\infty}^{z} f(z)\, dz = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = P[Z \le z]$$

NOTE (1) The cumulative probabilities $F(z)$ correspond to the area under the standard normal curve to the left of z. (shaded portion) figure (1)

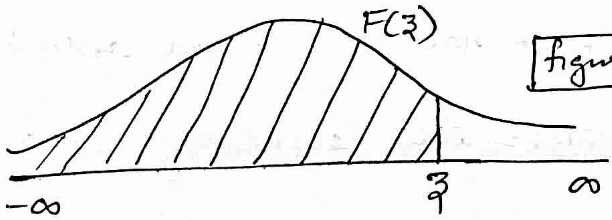
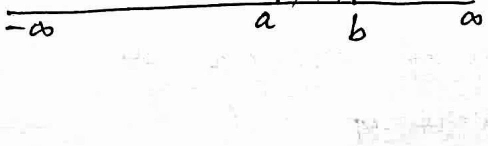


figure (1)

NOTE (2)

$P[a < z < b] = F(b) - F(a)$ = Area shown in figure (2)

figure (2)



NOTE (3)

Let X be a normal variable with mean μ and variance σ^2. Then we write as $X \sim N(\mu, \sigma^2)$. Also $Z = \frac{X-\mu}{\sigma}$ is a standard normal variable. Since,

Mean of $Z = E(Z) = E\left[\frac{X-\mu}{\sigma}\right] = \frac{1}{\sigma}\left[E(X) - E(\mu)\right] = \frac{1}{\sigma}(\mu - \mu) = 0$

Variance of $Z = Var\left[\frac{X-\mu}{\sigma}\right] = \frac{1}{\sigma^2}\left[V(X) - V(\mu)\right] = \frac{1}{\sigma^2}\left[\sigma^2 - 0\right] = 1$

[Q] Let X be a normal variable with mean $\mu = 4.35$ and $\sigma = 0.59$. Find (i) $P[4 < X < 5]$ (ii) $P[X > 5.5]$

Sol: Here, $X \sim N(\mu, \sigma^2)$ with $\mu = 4.35$, $\sigma = 0.59$

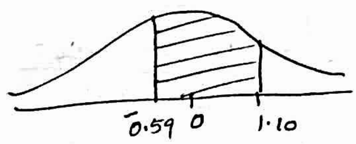
Now, $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$

(i) To find $P[4 < X < 5]$

$$P[4 < X < 5] = P\left[\frac{4-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{5-\mu}{\sigma}\right]$$

$$= P\left[\frac{4-4.35}{0.59} < Z < \frac{5-4.35}{0.59}\right]$$

$$= P\left[-0.59 < Z < 1.10\right]$$



$$= F(1.10) - F(-0.59)$$

$$= 0.8643 - 0.2776 \text{ (From table)}$$

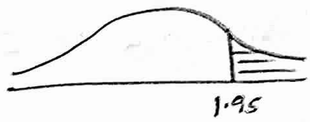
$$= \underline{\underline{0.5867}}$$

(ii) To find $P[x > 5.5]$

$$P[x > 5.5] = P\left[\frac{X-\mu}{\sigma} > \frac{5.5-\mu}{\sigma}\right]$$

$$= P\left[z > \frac{5.5 - 4.35}{0.59}\right]$$

$$= P[z > 1.95]$$



$$= 1 - F(1.95)$$

$$= 1 - 0.9744$$

$$= 0.0256$$

[Q] Thickness of a polyester film base has a mean 140 and standard deviation 2. Using normal distribution, find the probability that the thickness tested is between 138 and 143

Sol:

$P[138 < X < 143]$; where X represents thickness

Given: $\mu = 140$ (mean)
$\sigma = 2$ (S.D)

$$= P\left[\frac{138-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{143-\mu}{\sigma}\right]$$

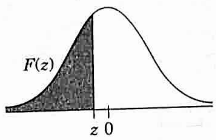
$$= P\left[\frac{138-140}{2} < z < \frac{143-140}{2}\right]$$

$$= P[-1 < z < 1.5]$$

$$= F(1.5) - F(-1) = 0.9332 - 0.1587 = 0.7745$$

Table III Standard Normal Distribution Function

$$F(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-t^2/2}\, dt$$



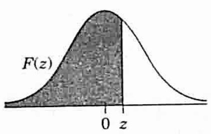
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−5.0	0.0000003									
−4.0	0.00003									
−3.5	0.0002									
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0006	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110

(*Continued*)

Table III (*Continued*)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

$$F(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-t^2/2}\, dt$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5973	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0,9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633

(*Continued*)

Table III *(Continued)*

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998									
4.0	0.99997									
5.0	0.9999997									

Q The weekly wages of 1000 workers are normally distributed around a mean of Rs. 70 and with a standard deviation of Rs. 5. Estimate the no. of workers whose weekly wages will be

(i) between Rs. 70 & Rs. 72

(ii) more than Rs. 75 (iii) less than 63.

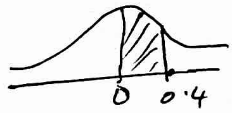
Sol: Given: $X \sim N(\mu, \sigma^2)$, $\mu = \text{mean} = 70$

$\sigma = S.D = 5$

(i) $P[70 < X < 72] = P\left[\frac{70-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{72-\mu}{\sigma}\right]$

$= P\left[\frac{70-70}{5} < Z < \frac{72-70}{5}\right]$

$= P[0 < Z < 0.4]$



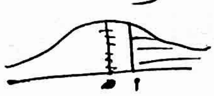
$= F[0.4] - F(0)$

$= 0.6554 - 0.5 = 0.1554$

$\therefore$ No. of workers with weekly wages between 70 & 72 is $= 1000 \times 0.1554 = 155.4 \equiv 155$

(ii) $P[X > 75] = P\left[\frac{X-\mu}{\sigma} > \frac{75-\mu}{\sigma}\right] = P\left[Z > \frac{75-70}{5}\right] = P[Z > 1]$

$= 1 - P[Z \leq 1]$



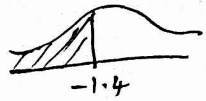
$= 1 - F(1)$

$= 1 - 0.8413 = 0.1587$

$\therefore$ No. of workers $= 1000 \times 0.1587 = 158.7 \equiv 159$

(iii) $P[X < 63] = P\left[Z < \frac{63-70}{5}\right] = P[Z < -1.4] = F[1.4]$

$= 0.0808$



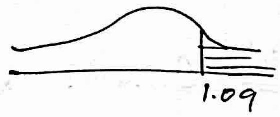
$\therefore$ No. of workers $= 1000 \times 0.0808 = 80.8 \simeq 81$

Q The marks obtained in Mathematics by 1000 students are normally distributed with mean = 78% and $\sigma = 11\%$. Determine the no. of students who got marks above 90%.

Sol: Given: $X \sim N(\mu, \sigma^2)$, $\mu = 78\%$, $\sigma = 11\%$

(Here: X-denote the marks obtained in Mathematics)

$$P[X > 90] = P\left[\frac{X-\mu}{\sigma} > \frac{90-\mu}{\sigma}\right]$$



$$= P\left[Z > \frac{90-78}{11}\right] = P[Z > 1.09] = 1 - P[Z < 1.09]$$

$$= 1 - F(1.09) = 1 - 0.8621 = 0.1379.$$

$\therefore$ No. of students who got marks above 90% $= 1000 \times 0.1379$

$= 137.9 \equiv \underline{\underline{138}}$

Q Let X be a normal variate with mean 20 and standard deviation 5. Find the probability that $P[|X-20| > 5]$

Sol: $P[|X-20| > 5] = P[(X-20) > 5 \text{ or } (X-20) < -5]$

$|x-20| > 5$

$\Rightarrow x-20 > 5$ (or)

$-(x-20) > 5$

$\Downarrow$

$(x-20) < -5$

Given

Mean = 20

$\Rightarrow \mu = 20$

S.D = 5

$\Rightarrow \sigma = 5$

$$= P[X > 25 \text{ or } X < 15]$$

$$= P[X > 25] + P[X < 15]$$

$$= P\left[\frac{X-\mu}{\sigma} > \frac{25-\mu}{\sigma}\right] + P\left[\frac{X-\mu}{\sigma} < \frac{15-\mu}{\sigma}\right]$$

$$= P\left[Z > \frac{25-20}{5}\right] + P\left[Z < \frac{15-20}{5}\right]$$

$$= \underbrace{P[Z > 1]} + P[Z < -1]$$

$$= 1 - P[Z \leq 1] + P[Z < -1]$$

$$= 1 - F(1) + F(-1)$$

$$= 1 - 0.8413 + 0.1587 = 0.3174$$

Q. A random variable has a normal distribution with $\mu = 62.4$. Find its standard deviation if the probability is 0.20 that it will take on a value greater than 79.2

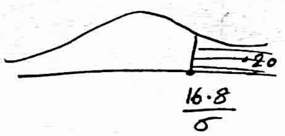
Sol: Given: $\mu = 62.4$, $\sigma = ?$

Also, given that $P[X > 79.2] = 0.20$

$$\Rightarrow P\left[\frac{X-\mu}{\sigma} > \frac{79.2-\mu}{\sigma}\right] = 0.20$$

$$\Rightarrow P\left[Z > \frac{79.2-62.4}{\sigma}\right] = 0.20$$

$$\Rightarrow P\left[Z > \frac{16.8}{\sigma}\right] = 0.20$$



$$\Rightarrow 1 - P\left[Z \le \frac{16.8}{\sigma}\right] = 0.20$$

$$\Rightarrow P\left[Z \le \frac{16.8}{\sigma}\right] = 0.80 \Rightarrow F\left[\frac{16.8}{\sigma}\right] = 0.80$$

From the table, $\frac{16.8}{\sigma} = 0.85 \Rightarrow \sigma = \frac{16.8}{0.85} = \frac{336}{17}$

Q. Assuming the mean height of soldiers to be 68.22 inches with a variance of 10.8, find how many soldiers in a regiment of 1000 would you expect to be over 6ft tall

Sol: Given: mean $= \mu = 68.22$, ~~SD~~ variance $= \sigma^2 = 10.8$

$P[\text{height greater than 6ft}] = P[X > 6ft] = P[X > 72 \text{ inches}]$

$$= P\left[\frac{X-\mu}{\sigma} > \frac{72-\mu}{\sigma}\right] = P\left[Z > \frac{72-68.22}{\sqrt{10.8}}\right]$$

Q. In a normal distribution, 17% of items are below 30 and 17% are above 60. Find mean & standard deviation

Sol: $X \sim$ Normal distribution

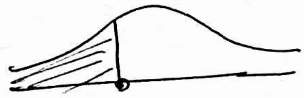
Given: $P[X<30] = 17\% = \frac{17}{100} = 0.17$ and

$P[X>60] = 17\% = \frac{17}{100} = 0.17$

we've to find μ & σ (ie mean and s.D).

$\boxed{\text{Now, } P[X<30] = 0.17}$

$\Rightarrow P\left[\frac{X-\mu}{\sigma} < \frac{30-\mu}{\sigma}\right] = 0.17$



$\Rightarrow P\left[Z < \frac{30-\mu}{\sigma}\right] = 0.17$

$Z_1 = \frac{30-\mu}{\sigma}$

$\Rightarrow P[Z < Z_1] = 0.17$ where $Z_1 = \frac{30-\mu}{\sigma}$

$\Rightarrow F(Z_1) = 0.17$; From table; $Z_1 = -0.95$

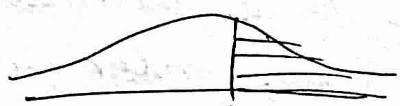
$\Rightarrow \frac{30-\mu}{\sigma} = -0.95$

$\Rightarrow 30-\mu = -0.95\sigma$

or $-\mu + 0.95\sigma = -30$ —— (1)

$\boxed{\text{Now, } P[X>60] = 0.17}$

$\Rightarrow P\left[\frac{X-\mu}{\sigma} > \frac{60-\mu}{\sigma}\right] = 0.17$



Z_2

$\Rightarrow P\left[Z > \frac{60-\mu}{\sigma}\right] = 0.17$

$\Rightarrow P[Z > Z_2] = 0.17$ $\Rightarrow 1 - P[Z<Z_2] = 0.17$

where $Z_2 = \frac{60-\mu}{\sigma}$ $\Rightarrow 1 - F(Z_2) = 0.17$

$1 - F(z_2) = 0.17$

$\Rightarrow \quad F(z_2) = 1 - 0.17 = 0.83$

$\Rightarrow \quad F(z_2) = 0.83 \quad \therefore$ From table, $z_2 = 0.96$

$$\Rightarrow \frac{60-\mu}{\sigma} = 0.96$$

$$\Rightarrow 60 - \mu = 0.96\sigma$$

$$\Rightarrow -\mu - 0.96\sigma = -60 \quad \text{—} (2)$$

From (1), $\quad -\mu + 0.95\sigma = -30$

From (2), $\quad -\mu - 0.96\sigma = -60$

Solving ; we get $\mu = 44.92$ and $\sigma = 15.706$

Home Work

Q) For a normally distributed population, 7% of the items have their values less than 35 and 89% have their values less than 63. Find the mean and S.D of the distribution.

Q) In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean & variance of the distribution.

[Q] 5% in a normal distribution are below 60 and 40% are between 60 & 65. Find mean and standard deviation.

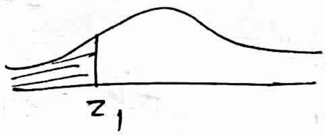
Sol: Given: ⊛ $P[X<60] = 5\% = \frac{5}{100} = 0.05$

⊛ $P[60 < X < 65] = 40\% = \frac{40}{100} = 0.40$

we've to find mean, 'μ' and S.D 'σ'.

⊛ $\boxed{P[X<60] = 0.05}$

$\Rightarrow P\left[\frac{X-\mu}{\sigma} < \frac{60-\mu}{\sigma}\right] = 0.05$



$\Rightarrow P[Z < Z_1] = 0.05, \quad \boxed{Z_1 = \frac{60-\mu}{\sigma}}$ —— (A)

$\Rightarrow \quad F(Z_1) = 0.05$ —— (B)

From, table : $Z_1 = -1.64$

$\therefore \quad \frac{60-\mu}{\sigma} = -1.64 \quad$ from (A)

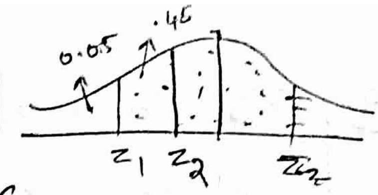
$\Rightarrow \quad 60 - \mu = -1.64\sigma$

$\Rightarrow \quad -\mu + 1.64\sigma = -60$ —— (1)

⊛ $P[60 < X < 65] = 0.4$

$\Rightarrow P\left[\underbrace{\frac{60-\mu}{\sigma}}_{z_1} < \frac{X-\mu}{\sigma} < \underbrace{\frac{65-\mu}{\sigma}}_{z_2}\right] = 0.4$

$P[z_1 < z < z_2] = 0.4$; $\boxed{z_2 = \frac{65-\mu}{\sigma}}$ → C



$\Rightarrow F(z_2) - F(z_1) = 0.4$

$\boxed{\because} = F(z_2) - F(z_1)$

$\Rightarrow F(z_2) - 0.05 = 0.4$

from (B)

$F(z_1) = \cancel{0.4}\ 0.05$

$\Rightarrow F(z_2) = 0.4 + 0.05 = 0.45$

From table, $z_2 = \bar{0}.12$

$\therefore$ ie $\frac{65-\mu}{\sigma} = \bar{0}.12$ from (C)

$\Rightarrow 65 - \mu = \bar{0}.12\sigma$

$\Rightarrow -\mu + 0.12\sigma = -65$ —— (2)

From (1); $-\mu + 1.64\sigma = -60$

From (2) $-\mu + 0.12\sigma = -65$

On solving, $\mu = 65.29$
$\sigma = 3.28$

Home work

Q On a large group of men, 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming normal distribution, find the average height and variance.

Joint Probability density function :

Let (X,Y) be a continuous random variable such that,

$$P\left[x < X \le x+dx \,,\; y < Y \le y+dy \right] = f(x,y)\,dx\,dy$$

Where dx and dy are small +ve real numbers. Then, $f(x,y)$ is called joint probability density function (pdf) of (X,Y).

Properties :

* $f(x,y) \geq 0 \quad \forall\, x, y$
* $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

Joint distribution function :

Let $f(x,y)$ be joint probability density function of (X,Y). Then joint distribution function $F(x,y)$ is given by,

$$F(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(x,y)\,dx\,dy \,.$$

NOTE:

$$f(x,y) = \frac{\partial^2 F}{\partial x\, \partial y}$$

[Q] Find the value of k if $f(x,y) = k(1-x)(1-y)$ where $0<x,y<1$ is the joint density function of (X,Y).

Sol: Since $f(x,y)$ is the joint pdf, we have

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$$

$$\Rightarrow \int_{y=0}^{1}\int_{x=0}^{1} f(x,y)\,dx\,dy = 1$$

$$\Rightarrow \int_{y=0}^{1}\int_{x=0}^{1} k(1-x)(1-y)\,dx\,dy = 1$$

$$\Rightarrow \int_{y=0}^{1} \left[k\left(x-\frac{x^2}{2}\right)\right]_{x=0}^{1}(1-y)\,dy = 1$$

$$\Rightarrow \int_{y=0}^{1} \left[k\left(1-\frac{1}{2}\right)-0\right](1-y)\,dy = 1$$

$$\Rightarrow \int_{y=0}^{1} \frac{k}{2}(1-y)\,dy = 1$$

$$\Rightarrow \frac{k}{2}\left[y-\frac{y^2}{2}\right]_0^1 = 1$$

$$\Rightarrow \frac{k}{2}\left[1-\frac{1}{2}\right] = 1 \quad \Rightarrow \frac{k}{2}\cdot\frac{1}{2} = 1 \quad \Rightarrow \frac{k}{4} = 1$$

$$\Rightarrow k = 4$$

Q If the joint pdf of (X,Y) is given by $f(x,y) = kxy\,e^{-(x^2+y^2)}$, $x>0,\ y>0$, then determine k.

Sol: Since $f(x,y)$ is the joint pdf, we've

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$$

$$\Rightarrow \int_{y=0}^{\infty}\int_{x=0}^{\infty} kxy\,e^{-(x^2+y^2)}\,dx\,dy = 1$$

$$\Rightarrow \int_{y=0}^{\infty}\int_{x=0}^{\infty} \left[k\,x\,y\,e^{-x^2}\cdot e^{-y^2}\right] dx\,dy = 1$$

(*) Since, All limits are constants & $f(x,y) = f(x)\,g(y)$. $\Rightarrow \int\int = \int \times \int$

(*)
$$\Rightarrow \int_{y=0}^{\infty} k\,y\,e^{-y^2}\,dy \cdot \int_{x=0}^{\infty} x\,e^{-x^2}\,dx = 1$$

put $y^2 = t$ $\Rightarrow 2y\,dy = dt$

put $x^2 = s$, $2x\,dx = ds$

$$\Rightarrow \int_{t=0}^{\infty} k\cdot e^{-t}\,\frac{dt}{2} \cdot \int_{s=0}^{\infty} e^{-s}\,\frac{ds}{2} = 1$$

$e^{-\infty} = 0$

$e^{-0} = 1$

$$\Rightarrow \frac{k}{2}\cdot\left(\frac{e^{-t}}{-1}\right)_{t=0}^{\infty} \cdot \frac{1}{2}\left(\frac{e^{-s}}{-1}\right)_{s=0}^{\infty} = 1$$

$$\Rightarrow \frac{-k}{2}\left[e^{-\infty} - e^{-0}\right] \cdot \frac{-1}{2}\left[e^{-\infty} - e^{-0}\right] = 1$$

$$\Rightarrow -\frac{k}{2}[0-1]\cdot -\frac{1}{2}[0-1] = 1 \Rightarrow \frac{k}{2}\cdot\frac{1}{2} = 1 \Rightarrow \frac{k}{4} = 1 \Rightarrow \underline{\underline{k=4}}$$

Q Find joint pdf $f(x,y)$ if joint distribution function is,

$$F(x,y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-x-y}, & x>0, y>0 \\ 0, & \text{otherwise} \end{cases}$$

Sol: we've : $f(x,y) = \dfrac{\partial^2 F}{\partial x \, \partial y}$

$$= \frac{\partial^2}{\partial x \, \partial y}\left[1 - e^{-x} - e^{-y} + e^{-x} \cdot e^{-y}\right]$$

$$= \frac{\partial}{\partial x}\left[0 - 0 - e^{-y} \times -1 + e^{-x} \cdot e^{-y} \times -1\right]$$

$$= \frac{\partial}{\partial x}\left[e^{-y} - e^{-x} \cdot e^{-y}\right]$$

$$= 0 - e^{-y} \cdot e^{-x} \cdot -1$$

$$= e^{-y} \cdot e^{-x} = e^{-x-y}$$

Thus, $f(x,y) = \begin{cases} e^{-x-y}, & x>0, y>0 \\ 0, & \text{otherwise.} \end{cases}$

Home Work

Q If $f(x,y) = c(xy^2 + e^x)$, $0<x<1$, $0<y<1$ is the joint pdf of (X,Y), then find the value of c.

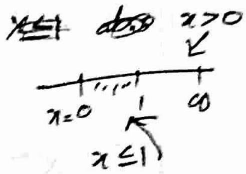
Ans : $\frac{6}{6e-5}$

Q If the joint density function of (X,Y) is

$$f(x,y) = \begin{cases} \frac{1}{2}, & x>0,\ y>0,\ x+y<2 \\ 0, & \text{otherwise}. \end{cases}$$ Then find

(i) $P[X\le 1, Y\le 1]$ (ii) $P[X+Y<1]$ (iii) $P[X>2Y]$

Sol: (i) To find $P[X\le 1, Y\le 1]$



$$P[X\le 1, Y\le 1] = \int_{y=0}^{1}\int_{x=0}^{1} f(x,y)\,dx\,dy$$

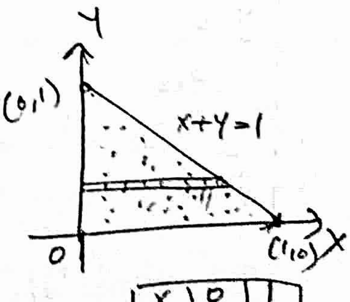
$$= \int_{y=0}^{1}\int_{x=0}^{1} \frac{1}{2}\,dx\,dy$$

$$= \int_{y=0}^{1} \frac{1}{2}[x]_0^1\,dy$$

$$= \int_{y=0}^{1} \frac{1}{2}\,dy = \frac{1}{2}(y)_0^1 = \underline{\underline{\frac{1}{2}}}$$

(ii) To find $P[X+Y\le 1]$

$\boxed{x+y<2}$



X	0	1
Y	1	0

$X+Y=1$

$$P[X+Y\le 1] = \iint f(x,y)\,dx\,dy$$

$$= \int_{y=0}^{1}\int_{x=0}^{1-y} \frac{1}{2}\,dx\,dy$$

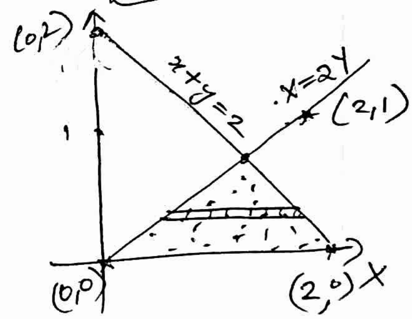
$$= \int_{y=0}^{1} \frac{1}{2}(x)\Big|_{x=0}^{1-y} dy$$

$$= \int_{y=0}^{1} \frac{1}{2}(1-y)\, dy = \frac{1}{2}\left[\left(y-\frac{y^2}{2}\right)\right]_{y=0}^{1} = \frac{1}{2}\left(1-\frac{1}{2}\right) = \underline{\underline{\frac{1}{4}}}$$

(iii) <u>To find $P[X>2Y]$</u>

$X=2Y$

x	0	2
y	0	1



$$P[X>2Y] = \iint f(x,y)\, dx\, dy$$

$$= \int_{y=0}^{2/3} \int_{x=2Y}^{2-y} \frac{1}{2}\, dx\, dy$$

$$= \int_{y=0}^{2/3} \left[\frac{1}{2} x\right]_{x=2y}^{2-y} dy$$

$$= \int_{y=0}^{2/3} \frac{1}{2}\left[(2-y) - (2y)\right] dy$$

$$= \int_{y=0}^{2/3} \frac{1}{2}\left[2 - y - 2y\right] dy$$

$$= \left[\frac{1}{2}\left(2y - \frac{y^2}{2} - 2\frac{y^2}{2}\right)\right]_{y=0}^{2/3}$$

$$= \frac{1}{2}\left[2\left(\frac{2}{3}\right) - \frac{1}{2}\left(\frac{4}{9}\right) - \frac{4}{9}\right] = \underline{\underline{\frac{1}{3}}}$$

given: $x+y<2$

$x+y=2$

x	0	2
y	2	0

<u>To get meeting pt. of $x+y=2$, $x=2y$</u>

$x+y=2 \Rightarrow 2y+y=2$

$\Rightarrow 3y=2$

$\Rightarrow y=2/3$

$x=2y$

$\therefore x = 2\times\frac{2}{3} = \frac{4}{3}$

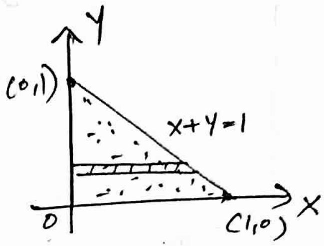
$\left(\frac{4}{3}, \frac{2}{3}\right)$

Q The joint pdf of a bivariate random variable X+Y is given by $f(x,y) = \begin{cases} 4xy, & 0<x<1,\ 0<y<1 \\ 0, & \text{otherwise} \end{cases}$. Find Then,

Find $P[X+Y<1]$

Sol:

$$P[X+Y<1] = \iint f(x,y)\,dx\,dy$$



X	0	1
Y	1	0

$$= \int_{y=0}^{1}\int_{x=0}^{1-y} 4xy\,dx\,dy$$

$$= \int_{y=0}^{1}\left[4y\frac{x^2}{2}\right]_{x=0}^{1-y} dy$$

$$= \int_{y=0}^{1} \frac{4y}{2}(1-y)^2\,dy$$

$(1-y)^2 = 1-2y+y^2$

$y(1-y)^2 = y-2y^2+y^3$

$$= 2\int_{y=0}^{1}(y-2y^2+y^3)\,dy$$

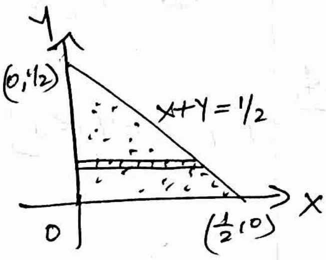
$$= 2\left[\frac{y^2}{2} - \frac{2y^3}{3} + \frac{y^4}{4}\right]_{y=0}^{1}$$

$$= 2\left[\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right]$$

$$= 2\left[\frac{3}{4} - \frac{2}{3}\right] = 2\left[\frac{9-8}{12}\right] = \frac{1}{6}$$

Q. If the joint pdf of X and Y is given by $f(x,y) = 24xy$ for $0<x<1$, $0<y<1$, $x+y<1$; then find $P[X+Y<\frac{1}{2}]$

Sol: $P[X+Y<\frac{1}{2}] = \iint f(x,y)\,dx\,dy$



X	0	1/2
Y	1/2	0

$$= \int_{y=0}^{1/2}\int_{x=0}^{1/2-y} 24xy\,dx\,dy$$

$$= \int_{y=0}^{1/2} 24y\left(\frac{x^2}{2}\right)_{x=0}^{\frac{1}{2}-y} dy$$

$$= \int_{y=0}^{1/2} \frac{24y}{2}\left(\frac{1}{2}-y\right)^2 dy$$

$$= \int_{y=0}^{1/2} 12y\left[\frac{1}{4}+y^2-y\right] dy$$

$$= \int_{y=0}^{1/2}\left(\frac{12y}{4} + 12y^3 - 12y^2\right) dy$$

$$= \left[3\frac{y^2}{2} + 12\cdot\frac{y^4}{4} - 12\frac{y^3}{3}\right]_{y=0}^{1/2}$$

$$= \frac{3}{2}\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right)^4 - 4\left(\frac{1}{2}\right)^3$$

$$= \frac{3}{8} + \frac{3}{16} - \frac{4}{8} = \frac{3}{16} - \frac{1}{8} = \frac{3}{16} - \frac{2}{16} = \underline{\underline{\frac{1}{16}}}$$

[9] Let X, Y be two continuous random variables with joint pdf $f(x,y) = \begin{cases} e^{-(x+y)}, & 0<x<\infty,\ 0<y<\infty \\ 0, & \text{otherwise} \end{cases}$

Find the distribution function of X and Y

Sol: Distribution function of X and Y is,

$$F(X,Y) = \int_{y=-\infty}^{y} \int_{x=-\infty}^{x} f(x,y)\,dx\,dy$$

$$= \int_{y=0}^{y} \int_{x=0}^{x} e^{-x} \cdot e^{-y}\,dx\,dy$$

$$= \int_{y=0}^{y} e^{-y} \cdot \left(\frac{e^{-x}}{-1}\right)_{x=0}^{x} dy$$

$$= \int_{y=0}^{y} -e^{-y} \cdot \left[e^{-x} - e^{-0}\right] dy$$

$$= \int_{y=0}^{y} -e^{-y}\left[e^{-x}-1\right] dy$$

$$= -\left[e^{-x}-1\right] \int_{y=0}^{y} e^{-y}\,dy$$

$$= -\left[e^{-x}-1\right]\left[\frac{e^{-y}}{-1}\right]_0^y = \underline{\underline{(e^{-x}-1)(e^{-y}-1)}}$$

Q If the joint pdf of X and Y is given by

$$f(x,y) = \begin{cases} e^{-(x+y)}, & x>0, y>0 \\ 0, & \text{otherwise} \end{cases}$$, then find

(i) $P[X<1]$ (ii) $P[X+Y<1]$

Sol: (i) To find $P[X<1]$

$$P[X<1] = \iint f(x,y)\, dA$$

$$= \int_{y=0}^{\infty}\int_{x=0}^{1} e^{-(x+y)}\, dx\, dy$$

$X<1 \Rightarrow$ no restriction for y. But $y>0$ is given.

Given: $f(x,y) = e^{-(x+y)}$ with $x>0,\ y>0$

$$= \int_{y=0}^{\infty}\left[\int_{x=0}^{1} e^{-x}\cdot e^{-y}\, dx\right] dy$$

$$= \int_{y=0}^{\infty} e^{-y}\left(\frac{e^{-x}}{-1}\right)_{x=0}^{1} dy$$

$$= \int_{y=0}^{\infty} -e^{-y}\left[e^{-1} - e^{-0}\right] dy$$

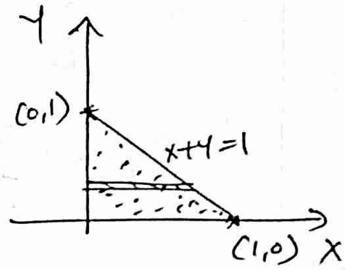
$$= \int_{y=0}^{\infty} -e^{-y}(e^{-1}-1)\, dy = (1-e^{-1})\left[\frac{e^{-y}}{-1}\right]_0^{\infty} = (e^{-1}-1)[0-1]$$

$$= 1-e^{-1}$$

(ii) To find $P[X+Y<1]$

$X+Y=1$

X	0	1
Y	1	0



$$P[X+Y<1] = \iint f(x,y)\,dA$$

$$= \int_{y=0}^{1} \int_{x=0}^{1-y} e^{-(x+y)}\,dx\,dy$$

$$= \int_{y=0}^{1} \left[\int_{x=0}^{1-y} e^{-x} \cdot e^{-y}\,dx \right] dy$$

$\left\{ \begin{array}{l} \text{Here : } f(x,y) = e^{-(x+y)} \\ \text{with } x>0,\ y>0 \end{array} \right.$

$$= \int_{y=0}^{1} \left[e^{-y} \cdot \frac{e^{-x}}{-1} \right]_{x=0}^{1-y} dy$$

$$= \int_{y=0}^{1} -e^{-y} \left[e^{-(1-y)} - e^{-0} \right] dy$$

$$= \int_{y=0}^{1} -e^{-y} \left[e^{-1+y} - 1 \right] dy$$

$$= \int_{y=0}^{1} \left(-e^{-y-1+y} + e^{-y} \right) dy$$

$$= \int_{y=0}^{1} \left(-e^{-1} + e^{-y} \right) dy = \left[-e^{-1}y + \frac{e^{-y}}{-1} \right]_{y=0}^{1} = \left(-e^{-1} + \frac{e^{-1}}{-1} \right) - \left(0 + \frac{1}{-1} \right)$$

$$= -e^{-1} - e^{-1} + 1 = \underline{1 - 2e^{-1}}$$

Marginal density function

Let X and Y be two continuous random variables with joint pdf $f(x,y)$. Then the

* Marginal density of $X = f_X(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy \quad , -\infty < x < \infty$

* Marginal density of $Y = f_Y(y) = \int_{x=-\infty}^{\infty} f(x,y)\,dx \,, -\infty < y < \infty$

Independent random variables:

Two continuous random variables X and Y are independent if and only if,

$$f(x,y) = f_X(x) \cdot f_Y(y) \quad \forall (x,y) \in R^2$$

where : $f(x,y)$ = joint pdf
$f_X(x)$ = marginal density fn. of X
$f_Y(y)$ = " " fn of Y.

NOTES:

* If X and Y are independent random variables then $E[XY] = E(X)\cdot E(Y)$; also

$$E[g(X)\cdot h(Y)] = E[g(X)] \cdot E[h(Y)]$$

* $E[X+Y] = E[X] + E(Y)$ & $E[aX+bY] = aE(X) + bE(Y)$

[Q] The joint pdf of (X,Y) is given by $f(x,y) = \begin{cases} kxy, & 0<x<4, \\ & 1<y<5 \\ 0, & \text{otherwise} \end{cases}$

(i) Find the value of k.

(ii) Determine marginal pdf of X and Y

(iii) Evaluate $P[X \geq 3, Y \leq 2]$

Sol: (i) To find K

Since $f(x,y)$ is a joint pdf, we have $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

$$\Rightarrow \int_{y=1}^{5}\int_{x=0}^{4} kxy\,dx\,dy = 1$$

$$\Rightarrow \int_{y=1}^{5}\left[ky \cdot \frac{x^2}{2}\right]_{x=0}^{4} dy = 1$$

$$\Rightarrow \int_{y=1}^{5} \frac{ky}{2}(16-0)\,dy = 1$$

$$\Rightarrow \int_{y=1}^{5} 8ky\,dy = 1$$

$$\Rightarrow 8k\left(\frac{y^2}{2}\right)_{y=1}^{5} = 1$$

$$\Rightarrow 4k\left[5^2 - 1^2\right] = 1 \Rightarrow 4k \times 24 = 1 \Rightarrow k = \frac{1}{96}$$

(ii) Marginal pdf of X and Y

(*) Marginal pdf of $X = f_X(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy$

ie $f_X(x) = \int_{y=-\infty}^{\infty} kxy\,dy$; here $k = \frac{1}{96}$ and $1<y<5$

$$= \int_{y=1}^{5} \frac{1}{96}\, xy\, dy$$

$$= \frac{1}{96}\, x \left(\frac{y^2}{2}\right)_{y=1}^{5} = \frac{1}{96}\frac{x}{2}\left(5^2-1^2\right) = \frac{x}{96\times 2}\times 24 = \frac{x}{8}$$

Thus, $f_X(x) = \frac{x}{8}$, $0<x<4$

(*) Marginal pdf of $Y = f_Y(y) = \int_{x=-\infty}^{\infty} f(x,y)\,dx$

ie $f_Y(y) = \int_{x=-\infty}^{\infty} kxy\,dx$; here $k = \frac{1}{96}$ and $0<x<4$

$$= \int_{x=0}^{4} \frac{1}{96}\, xy\, dx$$

$$= \frac{1}{96}\, y \left(\frac{x^2}{2}\right)_{0}^{4} = \frac{1}{96}\frac{y}{2}\left(4^2-0\right) = \frac{y}{12}$$

Thus, $f_Y(y) = \frac{y}{12}$, $1<y<5$

d

(iii) To find $P[X \geq 3, Y \leq 2]$

$$P[X \geq 3, Y \leq 2] = \iint f(x,y)\,dx\,dy$$

Given : $0 < x < 4$ & $1 < y < 5$

$$= \int_{y=1}^{2} \int_{x=3}^{4} \frac{1}{96} xy \, dx \, dy$$

$$= \int_{y=1}^{2} \frac{1}{96} y \left(\frac{x^2}{2}\right)_{x=3}^{4} dy$$

$$= \int_{y=1}^{2} \frac{1}{96} y \cdot \frac{1}{2} \cdot (16-9) \, dy$$

$$= \frac{1}{96} \cdot \frac{1}{2} \cdot 7 \left(\frac{y^2}{2}\right)_{y=1}^{2}$$

$$= \frac{7}{96 \times 2 \times 2} [4-1]$$

$$= \frac{21}{96 \times 4} = \frac{7}{128}.$$

Q Let X and Y be jointly distributed with pdf

$$f(x,y) = \begin{cases} \frac{1}{4}(1+xy), & |x|<1, |y|<1 \\ 0, & \text{otherwise} \end{cases}$$

Show that X and Y are not independent.

Sol:

$$f_X(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy = \int_{y=-1}^{1} \frac{1}{4}(1+xy)\,dy$$

$$= \left[\frac{1}{4}\left(y + x\frac{y^2}{2}\right)\right]_{y=-1}^{1} = \frac{1}{4}\left[\left(1+\frac{x}{2}\right) - \left(-1+\frac{x}{2}\right)\right]$$

$$= \frac{1}{4}\left[1+\frac{x}{2}+1-\frac{x}{2}\right] = \frac{1}{2}, \quad |x|<1$$

$$f_Y(y) = \int_{x=-\infty}^{\infty} f(x,y)\,dx = \int_{x=-1}^{1} \frac{1}{4}(1+xy)\,dx$$

$$= \left[\frac{1}{4}\left(x + y\frac{x^2}{2}\right)\right]_{x=-1}^{1} = \frac{1}{4}\left[\left(1+\frac{y}{2}\right) - \left(-1+\frac{y}{2}\right)\right]$$

$$= \frac{1}{4}\left[1+\frac{y}{2}+1-\frac{y}{2}\right] = \frac{1}{2}, \quad |y|<1$$

$$f_X(x) \cdot f_Y(y) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \neq f(x,y)$$

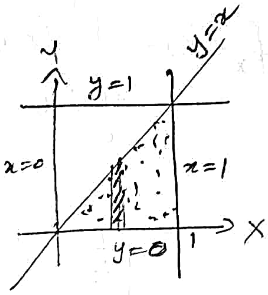
Thus, we showed that X and Y are not independent.

Q. The joint pdf of (X,Y) is $f(x,y)=\begin{cases} 8xy, & 0<y<x<1 \\ 0, & \text{otherwise} \end{cases}$

Examine the independence of X and Y.

Sol: X and Y are independent if $f(x,y)=f_X(x)\cdot f_Y(y)$.

Now, $f_X(x)=\int_{y=-\infty}^{\infty} f(x,y)\,dy \quad , -\infty<x<\infty$



$$=\int_{y=0}^{x} 8xy\,dy$$

$$=8x\left(\frac{y^2}{2}\right)_{y=0}^{x}$$

$$=\frac{8x}{2}\left(x^2-0\right) = \underline{\underline{4x^3}} \quad , \quad 0<x<1$$

Also, $f_Y(y)=\int_{x=-\infty}^{\infty} f(x,y)\,dx \quad , -\infty<y<\infty$

$$=\int_{x=y}^{1} 8xy\,dx$$

$$=8y\left(\frac{x^2}{2}\right)_{x=y}^{1}$$

$$=\underline{\underline{4y\left[1-y^2\right]}} \quad , \quad 0<y<1$$

$$f_X(x)\cdot f_Y(y)=4x^3 \times 4y(1-y^2) = 16x^3y(1-y^2)$$

$$\neq f(x,y) \Rightarrow \underline{\underline{X \text{ and } Y \text{ are not independent}}}$$

g

Q The joint pdf of X and Y is given by

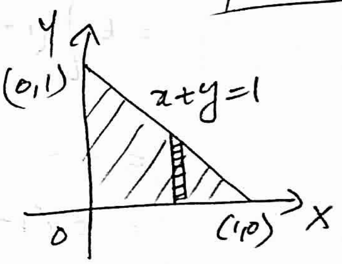
$$f(x,y) = \begin{cases} 6(1-x-y), & x>0,\ y>0,\ x+y<1 \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginal density functions of X and Y. Also examine whether X and Y are independent.

Sol:

$$\text{Mar. den. fn. of } X = f_X(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy, \quad -\infty < x < \infty$$

x	0	1
y	1	0



$$\Rightarrow f_X(x) = \int_{y=0}^{1-x} f(x,y)\,dy$$

$$= \int_{y=0}^{1-x} (6-6x-6y)\,dy$$

$$= \left[6y - 6xy - \frac{6y^2}{2}\right]_{y=0}^{1-x} = 6(1-x) - 6x(1-x) - 3(1-x)^2$$

$$= 6 - 6x - 6x + 6x^2 - 3(1-x)^2$$

$$= 6 - 12x + 6x^2 - 3(1-x)^2$$

$$= 6(1-2x+x^2) - 3(1-x)^2$$

$$= 6(1-x)^2 - 3(1-x)^2$$

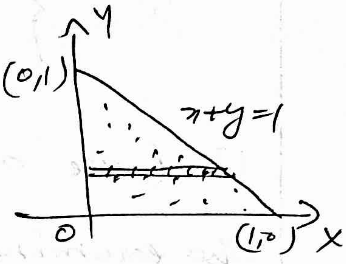
$$= 3(1-x)^2$$

Thus, $f_X(x) = 3(1-x)^2$; $0<x<1$

h

Marginal density fn of Y $= f_Y(y) = \int_{x=-\infty}^{\infty} f(x,y)\,dx \,, \quad -\infty < y < \infty$

ie $f_Y(y) = \int_{x=0}^{1-y} f(x,y)\,dy$



$$= \int_{x=0}^{1-y} 6(1-x-y)\,dx$$

$$= 6\left[x - \frac{x^2}{2} - yx\right]_{x=0}^{1-y}$$

$$= 6\left[(1-y) - \frac{(1-y)^2}{2} - y(1-y)\right]$$

$$= \ldots$$

$$= \underline{\underline{3(1-y)^2}}$$

Thus $\underline{\underline{f_Y(y) = 3(1-y)^2}}, \quad 0<y<1$

Now,

$$f_X(x) \cdot f_Y(y) = 3(1-x)^2 \cdot 3(1-y)^2$$

$$= 9(1-x)^2(1-y)^2$$

$$\neq f(x,y)$$

Hence X and Y are not independent random variables.

Q. Given the pdf : $f(x,y) = \begin{cases} \frac{2}{\pi}(1-x^2-y^2), & 0 < x^2+y^2 < 1 \\ 0, & \text{otherwise} \end{cases}$

(i) Find $f_X(x)$, (ii) Find $F_Y(y)$ (iii) $P[X^2+Y^2 < b^2]$, $0<b<1$

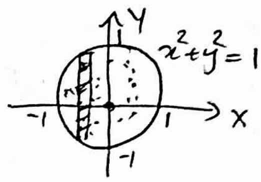
Sol: (i) $f_X(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy, \quad -\infty < x < \infty$

$\left\{ x^2+y^2=1 \Rightarrow \text{Circle with centre } (0,0), r=1 \right.$

$$= \int_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{\pi}(1-x^2-y^2)\,dy$$

$\downarrow$

even fn



$y^2 = 1-x^2$

$y = \pm\sqrt{1-x^2}$

$$\therefore = \int_{y=0}^{\sqrt{1-x^2}} \frac{4}{\pi}(1-x^2-y^2)\,dy$$

$$= \frac{4}{\pi}\left[y - x^2y - \frac{y^3}{3} \right]_{y=0}^{\sqrt{1-x^2}}$$

$\boxed{\sqrt{\ } \equiv 1/2 \quad (\sqrt{\ })^3 = \left(\frac{1}{2}\right)^3}$

$$= \frac{4}{\pi}\left\{ \sqrt{1-x^2} - x^2\sqrt{1-x^2} - \frac{1}{3}(1-x^2)^{3/2} \right\}$$

$$= \frac{4}{\pi}\left\{ \sqrt{1-x^2}\,(1-x^2) - \frac{1}{3}(1-x^2)^{3/2} \right\}$$

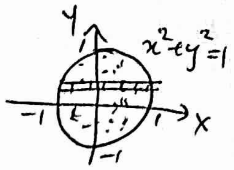
$$= \frac{4}{\pi}\left\{ (1-x^2)^{\frac{3}{2}} - \frac{1}{3}(1-x^2)^{3/2} \right\}$$

$\left\{ \begin{array}{l} * \sqrt{1-x^2} = (1-x^2)^{1/2} \\ \therefore \sqrt{1-x^2}\,(1-x^2) = (1-x^2)^{\frac{1}{2}+1} \\ \quad = (1-x^2)^{\frac{3}{2}} \end{array} \right.$

$$= \frac{4}{\pi}\left\{ (1-x^2)^{\frac{3}{2}} \left[1 - \frac{1}{3}\right] \right\}$$

$$= \frac{4}{\pi}\left\{ (1-x^2)^{3/2}\left(\frac{2}{3}\right) \right\} = \underline{\underline{\frac{8}{3\pi}(1-x^2)^{3/2}}}, \quad -1<x<1$$

(ii) $f_Y(y) = \int_{x=-\infty}^{\infty} f(x,y)\,dx = \int_{x=-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \frac{2}{\pi}(1-x^2-y^2)\,dx$



$x^2 = 1-y^2$

$x = \pm\sqrt{1-y^2}$

$$= 2\int_{x=0}^{\sqrt{1-y^2}} \frac{2}{\pi}(1-x^2-y^2)\,dx$$

$$= \frac{4}{\pi}\left[x - \frac{x^3}{3} - y^2 x\right]_{x=0}^{\sqrt{1-y^2}} = \frac{4}{\pi}\left\{\sqrt{1-y^2} - \frac{(1-y^2)^{3/2}}{3} - y^2\sqrt{1-y^2}\right\}$$

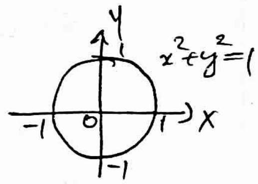
$$= \frac{4}{\pi}\left[\sqrt{1-y^2}(1-y^2) - \frac{1}{3}(1-y^2)^{3/2}\right]$$

$$= \frac{4}{\pi}\left[(1-y^2)^{3/2} - \frac{1}{3}(1-y^2)^{3/2}\right] = \frac{4}{\pi}\left[(1-y^2)^{3/2}\left(1-\frac{1}{3}\right)\right]$$

$$= \frac{4}{\pi}\left[(1-y^2)^{3/2}\cdot\frac{2}{3}\right] = \underline{\frac{8}{3\pi}(1-y^2)^{3/2}} \;;\; -1<y<1$$

(iii) $P[X^2+Y^2<b^2] = \iint f(x,y)\,dx\,dy.$

$$= \iint \frac{2}{\pi}(1-x^2-y^2)\,dx\,dy.$$



Given: $0<b<1$

$x^2+y^2<b^2$

Using polar coordinates:

$x = r\cos\theta,\; y = r\sin\theta,\; dx\,dy = r\,dr\,d\theta.$

$$= \int_{\theta=0}^{2\pi}\int_{r=0}^{b} \frac{2}{\pi}(1-r^2\cos^2\theta - r^2\sin^2\theta)\,r\,dr\,d\theta$$

$$= \int_{\theta=0}^{2\pi}\int_{r=0}^{b} \frac{2}{\pi}(1-r^2)\,r\,dr\,d\theta = \int_{\theta=0}^{2\pi}\left[\int_{r=0}^{b}\frac{2}{\pi}(r-r^3)\,dr\right]d\theta$$

$$= \int_{\theta=0}^{2\pi} \frac{2}{\pi}\left(\frac{r^2}{2}-\frac{r^4}{4}\right)_{r=0}^{b} d\theta = \int_{\theta=0}^{2\pi}\frac{2}{\pi}\left(\frac{b^2}{2}-\frac{b^4}{4}\right)d\theta = \frac{2}{\pi}\left(\frac{b^2}{2}-\frac{b^4}{4}\right)(\theta)_0^{2\pi} = \frac{2}{\pi}\left(\frac{b^2}{2}-\frac{b^4}{4}\right)2\pi$$

$$= 4\left(\frac{b^2}{2}-\frac{b^4}{4}\right) = \underline{\underline{2b^2-b^4}}$$

Q A gun is aimed at a certain point (origin of the co-ordinate system). Because of the random factors, the actual hit can be any point (X, Y) in a circle of radius R. Assume that the joint density of X and Y is constant in this circle given by,

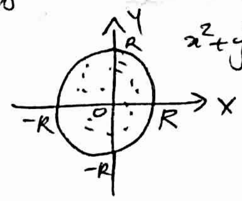
$$f(x,y) = \begin{cases} k, & x^2+y^2 \le R^2 \\ 0, & \text{otherwise} \end{cases}$$

Then (i) find k

(ii) find $f_X(x)$.

Sol: (i) To find k

Since $f(x,y)$ is a joint pdf, we've $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$



$x^2+y^2=R^2$

ie $\iint k\,dx\,dy = 1$

$\Rightarrow k \iint dx\,dy = 1$

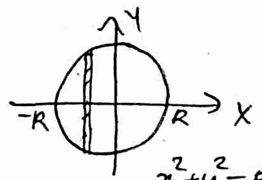
$\Rightarrow k\,[\text{area of circle } x^2+y^2=R^2] = 1$

$\Rightarrow k\,[\pi R^2] = 1$

$\Rightarrow k = \dfrac{1}{\pi R^2}$

(ii) To find $f_X(x)$

$$f_X(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy = \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} k\,dy$$



$x^2+y^2=R^2$

$y^2=R^2-x^2$

$y=\pm\sqrt{R^2-x^2}$

$$= k\,[y]_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} = k\left[\sqrt{R^2-x^2} - -\sqrt{R^2-x^2}\right]$$

$= k \times 2\sqrt{R^2-x^2}$; but $k = \dfrac{1}{\pi R^2}$

$= \dfrac{1}{\pi R^2} \times 2\sqrt{R^2-x^2}$; $-R < x < R$

Home Works

1] Evaluate C if the joint pdf of X and Y is

$$f(x,y) = \begin{cases} k(4-x-y), & 0 \le x \le 2,\ 0 \le y \le 2 \\ 0, & \text{Otherwise} \end{cases}$$

Also, examine whether X and Y are independent.

2] The joint pdf of a two-dimensional random variable (X,Y) is $f(x,y) = xy^2 + \frac{x^2}{8}$, $0 \le x \le 2$, $0 \le y \le 1$

Compute (i) $P[X>1]$ (iii) $P[X<Y]$

(ii) $P[Y<1/2]$ (iv) $P[X+Y \le 1]$

Expectation of multiple random variables

Let (X,Y) be a two-dimensional continuous random variable. Then expectation of $g(X,Y)$ is defined as

$$E[g(X,Y)] = \int_{y=-\infty}^{\infty}\int_{x=-\infty}^{\infty} g(x,y)\, f(x,y)\,dx\,dy$$

Q If X and Y are independent continuous random variables, then prove that $E(XY) = E(X)\cdot E(Y)$.
(Multiplication theorem of expectation).

Sol: $E(XY) = \int_{y=-\infty}^{\infty}\int_{x=-\infty}^{\infty} xy\, f(x,y)\,dx\,dy$

$$= \int_{y=-\infty}^{\infty}\int_{x=-\infty}^{\infty} xy\, f_X(x)\cdot f_Y(y)\,dx\,dy \qquad \left\{ \because X \,\&\, Y \text{ are indept; } f_X(x)\cdot f_Y(y) = f(x,y) \right.$$

$$= \int_{y=-\infty}^{\infty} y\, f_Y(y)\,dy \times \int_{x=-\infty}^{\infty} x f_X(x)\,dx$$

$$= E(Y)\cdot E(X)$$

$$= E(X)\cdot E(Y)$$

Q The joint pdf of the random variables X & Y is given by $f(x,y) = e^{-(x+y)}$, $x \geq 0$, $y \geq 0$. Find $E(XY)$.

Sol: $E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x,y)\,dx\,dy$

$$E(XY) = \int_{y=0}^{\infty}\int_{x=0}^{\infty} xy\, e^{-(x+y)}\, dx\, dy$$

$$= \int_{y=0}^{\infty}\int_{x=0}^{\infty} xy\, e^{-x} \cdot e^{-y}\, dx\, dy$$

$$= \int_{y=0}^{\infty} y e^{-y}\, dy \times \int_{x=0}^{\infty} x e^{-x}\, dx$$

$$\left\{ \begin{array}{l} u = y, \quad v = e^{-y} \\ u' = 1 \quad v_1 = \frac{e^{-y}}{-1} \\ u'' = 0 \\ \qquad v_2 = \frac{e^{-y}}{(-1)^2} \\ \int uv = uv_1 - u'v_2 + \cdots \end{array} \right.$$

$$= \left[y \cdot \frac{e^{-y}}{-1} - 1 \cdot e^{-y} \right]_{y=0}^{\infty} \times \left[x \frac{e^{-x}}{-1} - 1 \cdot e^{-x} \right]_{x=0}^{\infty}$$

$$= [0 - 0] - [0 - 1] \times [0 - 0] - [0 - 1]$$

$$= 1 \times 1 = \underline{\underline{1}}$$

[Q] If (X,Y) is a two-dimensional random variable with joint pdf, $f(x,y) = \begin{cases} 4xy, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$, find $E(XY)$.

<u>Sol:</u> $$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x,y)\, dx\, dy$$

$$= \int_{y=0}^{1}\int_{x=0}^{1} xy \cdot 4xy\, dx\, dy$$

$$= \int_{y=0}^{1} 4y^2\, dy \times \int_{x=0}^{1} x^2\, dx$$

$$= 4\left(\frac{y^3}{3}\right)_0^1 \times \left(\frac{x^3}{3}\right)_0^1 = \frac{4}{3} \times \frac{1}{3} = \underline{\underline{\frac{4}{9}}}$$

i.i.d random variables

A sequence of random variables $X_1, X_2, \cdots$ is identically and independently distributed (i.i.d) if they are all independent and all of them have the same probability distribution.

Central Limit theorem

If $X_1, X_2, \cdots X_n \cdots$ be a sequence of independent and identically distributed (i.i.d) random variables with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$, $i = 1, 2, \cdots$ and if $S_n = X_1 + X_2 + \cdots + X_n$, then S_n follows a normal distribution with 'mean $n\mu$' and 'variance $n\sigma^2$' as n tends to infinity.

Remark:

$$\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n) = \frac{1}{n}(S_n)$$

$$\Rightarrow E(\bar{X}) = E\left[\frac{1}{n} S_n\right] = \frac{1}{n} E(S_n) = \frac{1}{n}(n\mu) = \mu \quad \text{and}$$

$$V(\bar{X}) = V\left(\frac{1}{n} S_n\right) = \frac{1}{n^2} V(S_n) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}$$

$\therefore \bar{X}$ follows $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ as $n \to \infty$

$\boxed{\text{NOTE:}}$

* $S_n \sim N(n\mu, \sigma\sqrt{n})$

* $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

$N(\underset{\text{mean}}{\underbrace{-}}, \underset{\text{S.D}}{\underbrace{-}})$

Q The lifetime of a certain brand of an electric bulb may be considered as a random variable with mean 1200 h and standard deviation 250 h. Find the probability, using central limit theorem, that the average lifetime of 60 bulbs exceeds 1250 h

Sol: Let X_i represent the lifetime of the bulb.

Given: Mean = 1200 h , S.Dev. = 250 h

$\Rightarrow \mu = 1200$, $\sigma = 250$

Let $\overline{X}$ denote the mean lifetime of 60 bulbs.

We know, by central limit theorem,

$\overline{X}$ follows $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$; where $\overline{X}$ denote average life time

ie $\overline{X}$ follows $N\left(\underbrace{1200}_{\mu \text{ of } \overline{X}}, \underbrace{\frac{250}{\sqrt{60}}}_{\sigma \text{ of } \overline{X}}\right)$

Now; $P[\overline{X} > 1250] = ?$

$$P[\overline{X} > 1250] = P\left[\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} > \frac{1250 - \overset{\text{Mean of } \overline{X}}{\mu}}{\underbrace{\sigma/\sqrt{n}}_{\text{S.D of } \overline{X}}}\right]$$

$$= P\left[Z > \frac{1250 - 1200}{\frac{250}{\sqrt{60}}}\right]$$

μ = mean = 1200

σ = S.D = 250

for $\overline{X}$, σ is $\frac{\sigma}{\sqrt{n}}$

$$= P[Z > 1.55]$$

$$= 1 - P[Z \leq 1.55]$$

$$= 1 - F[1.55]$$

$$= 1 - 0.9394$$

$$= \underline{0.0606}$$

Q. If $X_1, X_2 \cdots X_n$ are Poisson variates with parameter $\lambda = 2$, use central limit theorem to estimate $P[120 \le S_n \le 160]$ where $S_n = X_1 + X_2 + \cdots + X_n$ and $n = 75$

Sol: Given : Poiss dis ; $\lambda = 2 \Rightarrow \begin{cases} \text{mean} = 2 \\ \mu = 2 \end{cases}$

also, for pois dist, $Var = \lambda \Rightarrow \begin{cases} \sigma^2 = \lambda \\ \Rightarrow \sigma^2 = 2 \Rightarrow \sigma = \sqrt{2} \end{cases}$

Poiss. dist.
$f(x) = \dfrac{e^{-\lambda} \cdot \lambda^x}{x!}, \lambda > 0$
parameter $= \lambda$
Mean = Variance = λ

By Central limit theorem,

S_n follows $N(n\mu, \sigma\sqrt{n})$

$\boxed{S_n \sim N(n\mu, \sigma\sqrt{n})}$
$n = 75$
$\mu = 2$
$\sigma = \sqrt{2}$

$\Rightarrow$ S_n follows $N(150, \sqrt{75}\sqrt{2})$

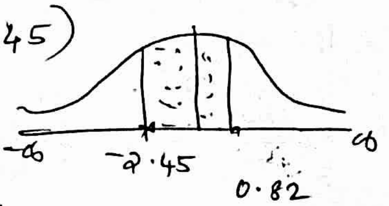
ie $S_n \sim N(\underbrace{150}_{\text{Mean of } S_n}, \underbrace{\sqrt{150}}_{\text{S.D of } S_n})$

$$P[120 \le S_n \le 160] = P\left[\frac{120-150}{\sqrt{150}} \le Z \le \frac{160-150}{\sqrt{150}}\right]$$

$$= P\left[\frac{-30}{\sqrt{150}} \le Z \le \frac{10}{\sqrt{150}}\right]$$

$$= P[-2.45 \le Z \le 0.82]$$

$$= F(0.82) - F(-2.45)$$



$= \cancel{0.8023} - \cancel{0.0071}$
$= 0.7939 - 0.0071$

$= \underline{0.7868}$

Q The life time of a certain brand of an electric bulb may be considered as an exponential random variable with mean 50 hours. Using central limit theorem, find the probability that 100 of these bulbs will provide a total of more than 6000 hours of burning time.

Sol.: Let X_i denote the life time of the bulb ; $X_i \sim$ exp. dis

Given : Mean $= 50 \Rightarrow \mu = 50$

Also : Variance $= 50^2 = 2500 \Rightarrow \sigma^2 = 2500$

$\Rightarrow \sigma = 50$

exp. dist.

$f(x) = e^{-\lambda x} \lambda$

mean $= E(X) = \frac{1}{\lambda}$

vari $= \frac{1}{\lambda^2}$

By central limit theorem,

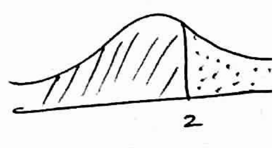
$S_n \sim N(n\mu, \sigma\sqrt{n})$; $n = 100$ given

ie $S_n \sim N(5000, 50\sqrt{100})$

$\Rightarrow S_n \sim N(5000, 500)$

Mean of S_n ; S.D of S_n

we're to find ; $P[S_n \geq 6000]$



////= F(2)

$\Rightarrow P\left[\frac{S_n - n\mu}{\sigma\sqrt{n}} \geq \frac{6000 - n\mu}{\sigma\sqrt{n}}\right]$

$\Rightarrow P\left[Z \geq \frac{6000 - 5000}{500}\right] = P\left[Z \geq \frac{1000}{500}\right] = P[Z \geq 2]$

$= 1 - P[Z < 2] = 1 - F(2)$

$= 1 - 0.9772$

$= \underline{0.0228}$

Q. A random sample of size 100 is taken from a population whose mean is 60 and variance is 400. Using Central limit theorem, with what probability can we assert that the mean of sample will not differ from $\mu = 60$ by more than 4.

Sol: Given: $n = 100$, $\mu = 60$, $\sigma^2 = 400$

{ sample size = 100
∴ Mean of sample
$= \bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{100}$ }

$P[|\bar{x} - 60| \le 4] = ?$

$P[|\bar{x} - 60| \le 4]$; converting to 'z'

$= P\left[\left|\frac{\bar{x} - 60}{\sigma/\sqrt{n}}\right| \le \frac{4}{\sigma/\sqrt{n}}\right]$

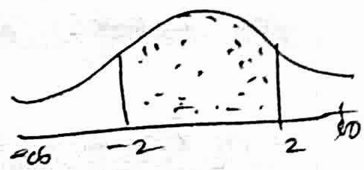
{ we know: $\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ }

$= P\left[|z| \le \frac{4}{2}\right]$

$\sigma^2 = 400$
$\Rightarrow \sigma = 20$
$n = 100$
$\sqrt{n} = 10$
$\frac{\sigma}{\sqrt{n}} = \frac{20}{10} = 2$

$= P[|z| \le 2] = P[-2 < z < 2]$

$= F(2) - F(-2)$



$= 0.9772 - 0.0228$

$= \underline{\underline{0.9544}}$

[Q] A computer generates 100 random numbers uniformly distributed between 0 and 1. Use 'CLT' to find the probability that (i) their sum is 60 or more
(ii) their average is 0.7 or less

Sol: Let X_i denote the number generated. Given that, ~~it is~~
$X_i \sim$ Uniform distribution. (b/n 0 and 1).

$\Rightarrow X_i \sim U(0,1) \quad \therefore \text{Mean} = \frac{0+1}{2} = \frac{1}{2} = \mu$

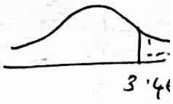
Also, given : $n = 100$ $\quad \text{Variance} = \frac{(1-0)^2}{12} = \frac{1}{12} = \sigma^2$

(*) (i) $P[S_n \geq 60] = ?$

we know: $S_n \sim N(n\mu, \sigma\sqrt{n})$ ie $S_n \sim N\left(100 \times \frac{1}{2}, \frac{1}{\sqrt{12}} \times \sqrt{100}\right)$

$\Rightarrow S_n \sim N\left(50, \frac{10}{\sqrt{12}}\right)$

Now, $P[S_n \geq 60] = P\left[\frac{S_n - n\mu}{\sigma\sqrt{n}} \geq \frac{60 - n\mu}{\sigma\sqrt{n}}\right]$



$= P\left[Z \geq \frac{60-50}{10/\sqrt{12}}\right] = P[Z \geq 3.46]$

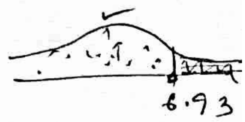
$= 1 - P[Z < 3.46] = 1 - 0.9997 = \underline{\underline{0.0003}}$

(*) (ii) $P[\bar{X} \leq 0.7] = ?$

we know: $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \Rightarrow \bar{X} \sim N\left(\frac{1}{2}, \frac{\frac{1}{\sqrt{12}}}{10}\right)$

$\therefore P[\bar{X} \leq 0.7] = P\left[\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{0.7 - \mu}{\sigma/\sqrt{n}}\right] = P\left[Z \leq \frac{0.7 - 1/2}{\frac{1}{\sqrt{12}}/10}\right]$

$= P[Z \leq 6.93]$



$= F[6.93]$

$= \underline{\underline{1}}$